Management and Performance in the Public Sector: Evidence from German Municipalities*

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October 26, 2026

*The paper is a substantial revision of a previous version circulated as CEPR Discussion Paper No. 17670, CESifo Working Paper No. 10060, and IZA Discussion Paper No. 15676. The project was preregistered at the Open Science Foundation (https://osf.io/3k7sb/). We gratefully acknowledge comments by Wouter Dessein, Luis Garicano, Ricard Gil, Anna Gumpert, Martin Hellwig, Renata Lemos, Imran Rasul, John van Reenen, Raffaella Sadun, Daniela Scur, Uwe Sunde, Joachim Winter, and Mu-Jeung Yang, from participants at the 2021 Meeting of the Committee for Organizational Economics of the German Economic Association, the 2022 SIOE conference (Toronto), the 2022 "Firms and Labor Workshop" (Holzhausen), the 2022 Empirical Management Conference (Washington D.C.), the 2025 Asian Conference on Organizational Economics (Hong Kong), the 2025 Workshop "The Public Administration and State Capacity" (Stuttgart), the 2026 Workshop "Managing the State" (Venice), as well as from seminar audiences at Bonn, Hamburg, Linz, Regensburg and University College Dublin. We thank Marte Anders, Anna Eisner, Lisa Eitinger, Pamela Fritzsch, Martina Kraus-Pietsch, Sebastian Kunz, and Leven Nendza for excellent research assistance, Dieter Daminger, Ulrich Hörning, and Siegfried Schloß for their input for the municipal survey, and the Bertelsmann Stiftung for kindly providing their data. We gratefully acknowledge support by Deutsche Forschungsgemeinschaft (CRC TRR 190, project 280092119). Finally, we thank the *Kommunale Gemeinschaftsstelle für Verwaltungsmanagement (KGSt)*, and in particular Marc Groß, Laura-Elena Diehl, Sophia Eich, Anika Krellmann, and Volker Scharfen, for their support in conducting our survey.

Abstract

This paper studies the management and performance of local government institutions. We conducted a management survey providing an anatomy of management practices employed in a representative sample of 600 municipal administrations in Germany. We find that municipalities differ widely in their use of management practices. Hence, while more heavily regulated than private firms, municipalities retain considerable leeway in their choice of management practices, with some utilizing this leeway while others do not. We then study the link between management and performance. We provide a broad set of indicators that, taken together, are informative regarding municipalities' performance relative to their (legally stipulated, but elusive) goal of promoting the well-being of their citizens. Using a management score as a measure of structured management (Bloom et al., 2019), we find systematic and robust evidence for a positive and economically significant relationship between municipalities' management score and their performance. Finally, using unsupervised machine learning (LDA) allows to classify municipalities' management style as either "administered" or "actively managed". We find that the latter style is infrequently used but associated with superior performance.

Keywords: Management practices, public sector organizations, local government, municipal performance, state capacity, World Management Survey (WMS)

JEL Classification: D20, D73, H11, H73, R50

1 Introduction

Motivation The public sector is a key actor in most advanced economies. For example, in 2023, general government spending in the United States, the United Kingdom, and Germany accounted for 39, 46, and 48 percent of GDP, respectively.[1] It is therefore not surprising that there exist a vast amount of economic research on the public sector. Thereby, the bulk of this literature has focused on optimal policy design (e.g., taxation or social security schemes).

Another strand of research on the public sector has placed state institutions at center-stage, stressing their key role for successful economic development (see e.g., Acemoglu and Robinson, 2012; Besley et al., 2022). The importance of capable state institutions has become particularly salient in the face of recent challenges such as COVID-19, global migration, and the green and digital transformations. It is therefore crucial to gain an understanding of the key factors determining state institutions' ability to cope with these challenges, and the effectiveness of public service provision in general. Thereby, one potentially crucial factor is how these institutions are managed. Starting with Bloom and Van Reenen (2007), a large body of research has studied in detail which specific management practices are employed in private and public sector organizations, and the possible link between management and performance.

While public services are provided at all layers of bureaucracy (e.g. supranational, federal, state, county, or municipal), local government institutions such as municipal administrations are generally the main point of contact for citizens with "the state". In Germany, for example, around 80 percent of all public services are provided at the municipal level (see e.g. Deutscher Beamtenbund, 2022). However, relatively little is known about how these local government institutions are actually managed.

Taken together, these observations motivate three research questions regarding the management of local government institutions. We do this in the context of municipalities in Germany, a G7 country with an almost proverbial bureaucratic tradition. German municipal administrations provide a large number of public services in domains such as fostering economic activity, local infrastructure (e.g. roads, water supply and sewage, child care, fire brigades), and engage in regular interactions with citizens and businesses (e.g. registry services, business licences). They account for roughly 30 percent of total public sector employment in Germany (see e.g., Ruge and Ritgen, 2021, p. 124).

The first research question concerns the use of *management practices* (Q1). In particular, we seek to gain a systematic understanding of the management practices that German municipal administrations actually employ (e.g. in areas such as HR policies, the monitoring of service quality, or process management). Next, we investigate the relationship between the management practices employed in municipal administrations and municipal performance. To this end, as a prerequisite the second research question relates to *performance measurement* (Q2). In particular, German state law assigns municipalities the goal of promoting the well-being of their citizens. For example, in the federal state of Bavaria, municipalities shall "create and sustain – within the boundaries of

[1] See e.g., https://data.oecd.org/gga/general-government-spending.htm#indicator-chart, accessed on 02-11-2026.

economic feasibility – the public facilities that are necessary for the economic, social, and cultural well-being of its citizens".[2] This rather elusive goal raises the question of how it can be operationalized and how performance relative to this goal can be measured. Finally, the third research question addresses the *management-performance link*, and hence the effectiveness of local government institutions (Q3).

Related literature In addressing the three research questions, we build on a sizeable and influential empirical literature on how firms are run (see e.g. Ichniowski et al. 1997, Bloom and van Reenen 2007, Bloom et al. 2012, 2015c and 2019, and the survey by Scur et al. 2021).[3] In particular, many studies are based on the *World Management Survey (WMS)* developed by Bloom and Van Reenen (2007), querying managers on their use of management practices in various domains (incentives, monitoring, operations, targets). The answers are aggregated in a *management score* at the establishment level, aiming to measure the degree of structured management, i.e. management practices that "are more specific, formal, frequent or explicit" (Bloom et al., 2019).[4] These studies document that (i) the way firms are managed can be meaningfully measured, (ii) there are persistent differences in the management practices employed, (iii) the management score is positively correlated with firm performance.[5]

The WMS approach has also been applied to study management practices in public-sector organizations (for a survey, see Scur et al., 2021). For example, a higher management score is associated with a lower mortality rate in hospitals (Bloom et al., 2015b, 2020), a higher SAT score in schools (Bloom et al., 2015a),[6] and a better teaching and research performance in universities (McCormack et al., 2014). In addition, various papers have studied the role of specific management practices such as autonomy, performance pay, and career concerns in more detail (see e.g., Bandiera et al., 2009; Duflo et al., 2012; Rasul and Rogger, 2018; Ashraf et al., 2020; Rasul et al., 2021; Leaver et al., 2021), many of them focussing on public-sector organizations in developing countries (for an overview, see Besley et al., 2022). Finally, a recent literature has studied the importance of managers of public sector organizations and their human capital (see e.g., Dwenger and Gumpert, 2023; Fenizia, 2022, 2024; Hjort et al., 2021; Decarolis et al., 2020; Carreri and Payson, 2021, 2024; Carreri, 2021).

Contribution Against this backdrop, the present paper studies the management and performance of local government institutions (in particular, municipal administrations), and it makes the following contributions: Regarding research question Q1, we developed a version of the World Management Survey (WMS) tailored to the institutional setting of German municipalities. In

[2]For example, see Article 57 of the Municipal Code of the Federal State of Bavaria, https://www.gesetze-bayern.de/Content/Document/BayGO/true (own translation). Similar provisions apply in all 16 German federal states.

[3]The study of the management of bureaucracies, and the role of private firms as a potential role model in this respect goes back to Max Weber (see Weber, 1922).

[4]For an overview of the by now extensive set of studies employing the WMS methodology, see also https://worldmanagementsurvey.org.

[5]Thereby, this literature also contributes to explaining persistent productivity differences even in narrowly defined industries (for an overview, see e.g., Syverson, 2011).

[6]See also Adelman et al. (2020) and Lemos et al. (2024).

collaboration with local government experts, we designed a survey to cover six management categories: goal setting, monitoring of service quality, organizational learning, structured processes, human resource management, and digitization. We targeted the top-level executives –mayors or heads of administration– of all German municipalities with more than 5,000 inhabitants, representing approximately 80% of the total population. A 19% response rate yielded 600 municipalities that are representative across a wide set of characteristics. Using the survey responses, we are the first to provide an anatomy of the management practices used in German local government. We find that municipalities are highly heterogeneous in terms of the management practices employed. For instance, constructing a management score for each municipality in the spirit of Bloom et al. (2019) as discussed above, the difference between the top and bottom quartiles of the score distribution is approximately 1.5 standard deviations. Importantly, this heterogeneity persists *within* states, despite municipalities being primarily regulated at the state level. This suggests that municipalities indeed possess considerable leeway in their choice of management practices, with some utilizing this leeway while others do not. Our analysis of *management styles* – defined as the systematic combination of specific practices – further supports these findings. Following Bandiera et al. (2020), we employ unsupervised machine learning (in particular, the LDA method due to Blei et al. 2003). We find that the bulk of municipalities adopt a passive, *administered* style, employing structured management only sparingly. In contrast, *actively managed* municipalities utilize such practices intensively across all six surveyed management categories. Finally, a tentative comparison suggests that municipalities use structured management practices less intensively than private manufacturing firms in Germany.

Regarding research question Q2, we measure municipal performance relative to the rather elusive goal of fostering citizens' well-being by a broad set of indicators from various domains. We assume that municipalities that systematically outperform others across these dimensions do better in achieving their statutory goal. These indicators are taken from a unique data set (*Wegweiser Kommune*) and pertain to information on municipalities' financial health and their attractiveness for citizens and businesses (e.g. relating to income, profits, migration, and infrastructure). Focussing solely on a single performance indicator could be misleading because municipalities potentially pursue different strategies to achieve their goals, where some might prioritize either infrastructure development or attracting businesses, or service quality (instead of pursuing all of them at the same time).

Regarding research question Q3, we provide systematic and robust evidence for a positive and economically significant relationship between municipalities' management score and their performance. This finding is robust when controlling for municipal characteristics and to a host of different specifications. Among the six surveyed management categories, it is mainly practices related to the monitoring of service quality and human resource management that are associated with higher municipal performance. Moreover, using the classification of municipalities derived from the LDA, we find that the *actively managed* style is associated with superior municipal performance relative to the passive, *administered* style.

As a further contribution, our paper is the first to identify a Germany-wide, task-specific

measure of municipal performance, the scope of digital service provision. This information has recently made available in the course of an initiative by the federal government to boost the online provision of public services at the federal, state, and municipal levels (where Germany is lagging behind). Also for this specific task, we find a robust positive relationship between the management score and municipal performance.

In sum, we complement the WMS literature by providing systematic and granular evidence on the set of management practices employed in Germany's municipal administrations, and the high degree of heterogeneity in this respect. Our findings suggest that how these local government institutions are managed might play an important role in determining their ability to cope with the challenges ahead.

The remainder of the paper is structured as follows: Section 2 provides information on the institutional background. The management survey is discussed in Section 3 (design and implementation) and Section 4 (questions, responses, management score, and heterogeneity across municipalities). Section 5 explains the objectives of municipalities and the performance measures used in the empirical analysis. Section 6 studies the relationship between municipal performance and the management score (and subscores thereof), and it also addresses the question of management styles. Section 7 discusses performance differences in a specific municipal task, the scope of digital service provision. Section 8 concludes. Appendix A contains a translation of the survey (originally in German) and gives detailed evidence on all survey questions. Appendix B contains additional tables and figures.

2 Institutional background

Germany is a federal state and broadly structured as follows: Below the federal level, there are 16 states (*Länder*). As of 2021, these are partitioned into 294 counties (*Landkreise*), which in turn are partitioned into a total of 10,799 municipalities (*Kommunen*).[7]

Counties and municipalities are subject to federal and state law, but the German constitution grants them strong rights of self-government. For example, according to Article 28 of the constitution, "municipalities must be guaranteed the right to regulate all local affairs on their own responsibility within the limits prescribed by the laws". As a result, around 80 percent of direct administrative services for citizens and businesses are provided at the local level (see e.g., Deutscher Beamtenbund, 2022), and they account for around 30 percent of total public sector employment and one third of total public expenditure in Germany (see e.g., Ruge and Ritgen, 2021, p. 124f). In particular, municipal administrations are tasked with the development and maintenance of local infrastructure (e.g., local roads, land-use plans, and development plans to promote the local economy), and public facilities (e.g., parks, sports facilities), provision of child care facilities (e.g., public nurseries and kindergartens), and maintaining registry services (e.g., ID and civil status

[7]The different layers of public administration in Germany are described in more detail by Ruge and Ritgen (2021) and Schrapper (2021). Some states have an additional administrative layer formed by groups of counties (*e.g.*, *Regierungsbezirke*).

registration).[8]

With respect to autonomy, German municipalities face three task types. First, some tasks are mandatory and provided on behalf of the state or the federal level (e.g., registry services, support for federal or state elections). A second task type is also mandatory, but municipalities have more autonomy in deciding how to fulfil their obligations (e.g. provision of public child care or fire brigades). Finally, some services are provided on a voluntary basis, and municipalities are free to decide on how strongly to engage in such activities if at all (e.g. sports and cultural facilities, or projects to promote the local economy). Overall, German municipalities are much less flexible than private sector organizations. For example, they cannot simply shut down the provision of mandatory services. Moreover, as public sector organizations they face stricter regulations, for example with respect to HR policies (e.g. dismissals are generally not possible, and there is less flexibility regarding remuneration). However, as we show below, municipalities still retain considerable leeway in their choice of management policies, and they strongly differ with respect to how intensely this leeway is used.

With respect to the governance structure in municipalities, most states stipulate a dual structure, with a *representative body* of (elected) council members on the one side facing an *executive body* represented by the (elected) mayor on the other (see e.g., Ruge and Rittgen, 2021, pp. 134.) The council holds the key management and control rights, while the mayor, in the role as *chief administrative officer*, is responsible for executing the decisions taken by the council and for managing all administrative matters.

3 Survey design and implementation

This section discusses in detail the design and implementation of our municipal management survey (see Sections 3.1 and 3.2, respectively).

3.1 Design

Our survey elicits the prevalence of various management practices in German municipalities. Thereby, we build on the literature on the *World Management Survey* (WMS). Initially, WMS studies were conducted as semi-structured phone surveys, but later studies have also employed closed-form versions of the WMS (see e.g. Bloom et al., 2019), and our paper also adopts this approach.[9]

While there is a large related literature that investigates management practices in the private sector, these studies do not provide an off-the-shelf template for doing so in local government in-

[8] Note that in contrast to the U.S., German municipalities are, in general, not responsible for schools and the local police force, which are mainly administered at the state level (with some responsibilities at the county level). For a more detailed description of municipal tasks, see Ruge and Ritgen (2021, p. 129). Counties are mainly responsible for services in areas such as social welfare, unemployment benefits, public health, garbage collection, immigration, motor vehicle registration, and building permits. At the time when our study was conducted, there were 107 (typically larger) municipalities in Germany that are not part of a county (*Kreisfreie Städte*). In these cases, all of the services usually provided by the relevant county are provided by the administration of the respective municipality as well.

[9] See for example the *Management and Organizational Practices Survey* (MOPS), which is a closed form version of the *World Management Survey* implemented by the U.S. Census Bureau, see e.g., Buffington et al. (2017).

stitutions. Rather, when devising the questionnaire for our setting, we needed to take into account both the similarities and differences between private and public sector organizations. On the one hand, both when running a manufacturing firm and a municipal administration of equal size, one must set strategies, ensure a smooth operation of the day-to-day business, and recruit qualified employees as well as motivate them to do a decent job. On the other hand, municipalities have a more limited choice concerning which goods and services to provide, and their primary aim is not profit maximization. Moreover, German public-sector labor law stipulates that municipalities have only limited scope for using standard human-resource management tools for incentivizing employees, such as incentive pay, the threat of dismissal, or perks such as company cars. Moreover, while there exist management surveys for (public) entities such as hospitals and schools, they are not readily applicable to our context of German municipal administrations. Hence, we have adapted established surveys as explained in the following.

For the design and the implementation of the survey, we have cooperated with the *Kommunale Gemeinschaftsstelle für Verwaltungsmanagement* (KGSt), a non-profit organization, that specializes in consulting services for the administrations of its member municipalities.[10] KGSt have extensive expertise when it comes to the institutional background of public administration in the federal structure of Germany, specific managerial aspects, and challenges in the public sector.

To design our (closed-form) survey, we started with a large set of potential questions (130+) from all versions of the *World Management Survey*, from which we selected questions and adapted them to fit our context. Thereby, we consulted with KGSt as well as with several active and former mayors and other executives of municipal administrations to identify management practices that are potentially important for public-sector performance.

In the resulting questionnaire we ask municipal executives (i.e. the mayor or, in large municipalities, the head of the municipal administration) about their management practices in six categories: (i) goal setting, (ii) monitoring of service quality, (iii) organizational learning, (iv) structured processes, (v) human resource management, and (vi) digitization. Specific questions and responses are discussed in more detail in Section 4.1. With respect to (ii), as municipalities are legally required to record a variety of financial performance measures, in our survey we explicitly ask about the tracking of *service quality*, which is optional.

The answer categories for most of the questions are either binary (yes/no) or on a 5-point Likert scale. The questionnaire also contains some free-form questions. As explained in more detail in Section 4.2 below, we use the answers from a total of 52 closed-form questions and subquestions to construct a *management score* in the spirit of Bloom et al. (2019).

3.2 Implementation and participation

The municipal survey was conducted online in April and May 2021. It was programmed in LimeSurvey, and all data was directly stored on a server of the University of Hamburg.

KGSt supported us logistically by sending out the invitation emails for the survey. Because municipal executives regularly receive invitations to participate in KGSt surveys, we were expect-

[10] Membership is voluntary and provides several benefits, but there is also a fee.

ing a positive effect on the response rate from using this channel. In the communication with the municipalities regarding our survey, KGSt informed them that it was carried out together with us as part of a scientific project.[11]

We restricted the sample to sufficiently large municipalities for two reasons: First, very small municipalities often do not have an administration of their own. Instead, they frequently join forces with other small neighboring municipalities to form an administrative entity (*Verbandsgemeinde*), the administration of which is located in the main municipality. In the other municipalities of such entities, the local "administration" often consists solely of a mayor who holds this office as an honorary side job, possibly supported by a (part-time) clerk. Second, research questions Q2 and Q3 address the issue of municipal performance, drawing on a rich set of performance measures that generally is only available for municipalities with more than 5,000 inhabitants (for details, see Section 5.2 below). In total, we contacted all 3,083 municipalities covered by these data.

Out of the 3,083 municipalities, 1,602 are KGSt members, and KGSt had the contact details of the mayor or other top executive in their database. For the 1,481 remaining municipalities we have hand-collected the respective contact details from municipalities' websites, and sent out personalized invitations again to the mayor or the top executive in the administration. To keep the channel of communication constant, also these invitations were sent out by KGSt.

In our analysis, we use those 600 out of the 3,083 municipalities contacted that have responded and completed at least 32 out of the 52 questions (including sub-questions) relevant for our management score.[12] This corresponds to a response rate of 19.5 percent, and these 600 participating municipalities cover approximately 20 percent of the German population.[13] Figure 1 provides an overview of the geographical distribution of the municipalities we approached (left panel) and the municipalities that participated (right panel). Both panels display broadly similar geographical patterns. As discussed below, our sample is also representative with respect to municipalities' population and regional type as well as their performance (see Sections 4.2 and 5.2, respectively).

4 An anatomy of management practices employed by German municipal administrations

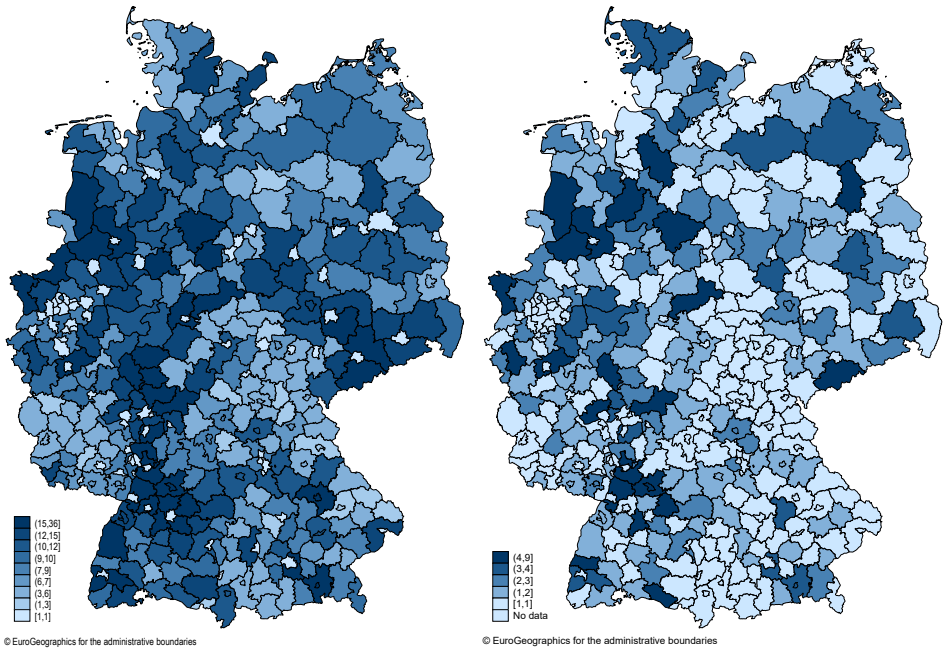
This section addresses research question Q1, i.e. we provide evidence on the use of various management practices in German municipalities. Section 4.1 discusses the survey questions and the respective responses. In Section 4.2, we calculate a summary management score for each municipality (in the spirit of Bloom et al., 2019), uncovering substantial heterogeneity even within classes

[11] To further incentivize participation, the invitation emails stated that 20 out of the responding municipalities would be randomly selected and invited to participate in an exclusive workshop "Challenges and Impulses for Public Management", which we have conducted jointly with KGSt in February 2022.

[12] In setting this threshold of 62 percent (32 out of 52 questions), we follow Bloom et al. (2019) who require at least 10 out of their 16 questions to be completed (63 percent). An additional 86 municipalities have participated in our survey, none of which has answered more than 27 questions, and 75% of these municipalities have answered at most 8 questions. These 86 municipalities were excluded from the analysis.

[13] The response rates for KGSt member and non-member municipalities are 21.8 and 16.9 percent, respectively.

Figure 1: Geographical distribution of invited municipalities (left panel) and participating municipalities (right panel) aggregated at the county level



Notes: Both panels show Germany partitioned into counties. In the left panel, the color indicates the number of municipalities with a population larger than 5,000 in the respective county (all of which were invited to the survey). In the right panel, the color indicates the number of municipalities in each county that have participated in the survey. To preserve anonymity, counties in which zero or one municipality participated are marked in the same color.

of comparable municipalities. Appendix A contains a translation of the questionnaire (originally in German) and detailed evidence on all surveyed management practices.

4.1 Survey questions and descriptive evidence

As discussed in Section 3.1, our survey questions can be grouped in six categories: First, for the management category "goal setting", we asked municipalities whether they have easily accessible mission statements directed either at citizens or at employees, whether these mission statements postulate goals, and about the time horizon of these goals. It turns out that goal setting through mission statements is rather rare: Around 80 percent of municipalities do not have any mission statement directed at citizens. Around 20 percent of existing mission statements do not formulate any goals. In those mission statements that do contain goals, more than 90 percent of goals are rather long-term, pertaining to a time horizon of more than five years. A similar picture emerges with respect to mission statements directed at employees, where the time horizon of goals is, however, somewhat shorter.

Second, with respect to the management category "monitoring of service quality", we inquired (i) whether applicable key performance indicators (such as processing times, waiting times, frequency of complaints) are tracked, (ii) whether customer satisfaction is elicited, and (iii) about the role such measures play for municipal management. The responses document that measuring

performance with respect to service quality is rare. Only 20 percent of municipalities use such indicators. If they do use them, the key indicator is processing times, while tracking waiting times or the frequency of complaints are less prevalent. Similarly, only around 15 percent of municipalities elicit customer satisfaction. Insights obtained from such performance tracking are usually discussed at the executive level of the administration (in around 90 percent of cases), and to a lesser degree also at the employee level (in around 60 percent of cases). Nevertheless, even if performance indicators are used, around 85 percent of the respective municipalities report that this has at most an intermediate impact on municipal management.

Third, as for the category "organizational learning", we focussed on (i) whether there are formal processes and incentives for employees to make suggestions, (ii) the impact of such suggestions on municipal management, and (iii) the sharing of management practices. In fact, 20 percent of municipalities provide incentives for employee suggestions, but in 70 percent of these municipalities, the suggestions have at most an intermediate impact on management. In about half of the municipalities there is a regular exchange within the administration or with other municipalities about management practices.

Fourth, as for the category "structured processes", we asked about whether tools such as project or process management are used. More than half of municipalities use project management, and these municipalities attach a high value to this tool. Process management is only employed by a minority of municipalities.

Fifth, as for the category "human resource management", there are legal restrictions on the use of (monetary) incentives in the German public sector (which is also the case in many other countries). Nevertheless, municipalities do have some leeway with respect to performance management and work organization, and our respective questions aimed to elicit to which degree this leeway is used. 78 percent of municipalities report to use "performance-based" remuneration of employees, which, however, reflects actual performance differences only to a small degree. For example, 73 percent of municipalities considered the link as either "very weak", "rather weak" or "intermediate", while only 27 percent consider it "strong" or "very strong". Municipalities use non-monetary rewards for good performance, in particular promotions (42 percent) and the possibility to participate in training programs (54 percent).[14] To attract high-skilled labor (a major challenge in the public sector), only a small fraction of municipalities offers compensation packages above collective bargaining agreement levels (15 percent), while flexible work hours (83 percent), and perks such as smartphones for office use (42 percent) are more common.

Finally, with respect to the category "digitization", for the areas of human resource management, financial management, and procurement processes we asked (i) which fraction of data is stored digitally, and (ii) to which degree data-driven forecasting is employed. The idea is that digitized data is easily accessible, lends itself to straightforward analysis, and can serve as the basis for data-driven decision making. It turns out that the use of such tools is limited. Even for an area such as financial management a substantial fraction of municipalities reports that only part of the

[14] We find that work from home was not very prevalent, which was mainly attributed to an inadequate IT infrastructure, data security concerns, impaired communication, but also due to concerns about unequal treatment of employees. Note that we explicitly asked about the pre-COVID-19 prevalence of such arrangements.

data is stored digitally. Moreover, few municipalities use their data to make projections, e.g. to gauge the future demand for child care.

To summarize, the survey responses document that German municipalities differ strongly in terms of their use of different management practices. Some make use of tools such as performance measurement or use existing leeway with respect to HR policies, while others do not. Again, see Appendix A for detailed results. As shown next, this heterogeneity is preserved when summarizing each municipality's survey responses in a single indicator, the *management score*.

4.2 Management score

In line with the literature on the *World Management Survey*, for each municipality we calculate a *management score*. Following Bloom et al. (2019), we use this measure to compactly capture the degree of structured management, i.e. whether management practices are "more specific, formal, frequent or explicit".[15] We find large heterogeneity in management scores. Moreover, this heterogeneity persists among municipalities within the same federal state, of similar size, or of the same regional type.

The management score is constructed as follows: The survey consists of a total of 49 questions, where 42 are directly linked to management practices.[16] Out of these 42 questions, 39 do not contain sub-questions, while questions Q30, Q45, and Q46 contain 3, 5, and 5 sub-questions, respectively. This leads to a total of $39 + 3 + 5 + 5 = 52$ items entering the management score. In a first step, we normalize the responses to each of the 52 management questions on a 0-1 scale, where 0 (1) indicates the least (most) "structured" management practice. For questions with more than two possible answers, we assign equidistant values between 0 and 1. For example, for a question with five possible answers, we assign values of 0, 0.25, 0.5, 0.75 and 1. In Appendix A, we detail for each management question the number of answer options and which of them is assigned the lowest value of 0. In a second step, we calculate the *management score* as the unweighed average of the normalized responses to each of the management questions. By construction, the value of the score lies between zero and one, and higher values indicate a higher degree of adoption of structured management practices.

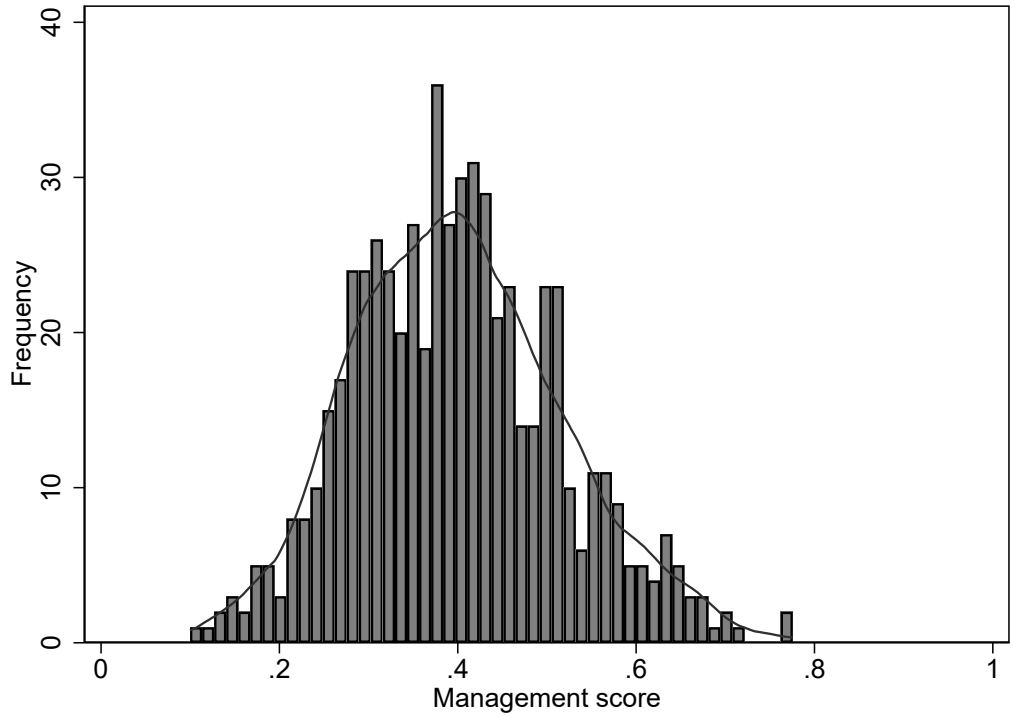
Figure 2 displays the frequency distribution of the management score documenting considerable heterogeneity. For example, the management score of municipalities at the 75% percentile is almost 1.5 standard deviations higher compared to the one at the 25% percentile. Given the relatively high degree of regulation in the public sector, this variation in the use of management practices seems remarkable.

German municipalities are mainly regulated at the state level. Hence, it could be that the observed heterogeneity in Figure 2 is mainly driven by *across-state* variation in regulations. Likewise, municipalities of similar size or regional type arguably operate in comparable environments.

[15] See p.2 of their online appendix.

[16] In the remaining seven questions we elicit (i) information regarding the relationship of the municipal administration with the local parliament (Q28), (ii) descriptive information about the respective municipal administration (Q29, Q42, Q43), and (iii) respondents' opinions about future challenges for municipal management (free-form questions, Q47, Q48, Q49, discussed in Section 6.2).

Figure 2: Frequency distribution of the management score



Notes: $N = 600$. Mean score: .40, standard deviation: .12, 25%-percentile: .31, 50%-percentile: .39, 75%-percentile: .48. The black line indicates the corresponding (Epanechnikov) kernel density.

This could lead them to employing similar management practices, yielding similar management scores. As for the definition of population size and regional type, we use classifications by the German Federal Statistical Office (reference year 2019). According to the *RegioStaR* classification (see Destatis, 2021, p.16ff), we distinguish four regional types of municipalities: (i) metropolitan, (ii) urban, (iii) rural close to urban, and (iv) rural.[17] With respect to population size, we use population size brackets based on the GKBIK7 categorization (see Destatis, 2021, p.44): (i) 2,001 - 5,000, (ii) 5,001 - 20,000, (iii) 20,001 - 50,000, (iv) 50,001 - 100,000, (v) 100,001 - 500,000, and (vi) more than 500,000 inhabitants.[18] Table 8 in Appendix B.1 displays the distributions of regional types and population size brackets, separately for participating and non-participating municipalities (and Kolmogorov-Smirnov tests reveal that these distributions do not differ significantly, i.e. our sample is representative along these dimensions).

Figure 3 documents that the heterogeneity of management scores observed in Figure 2 is not driven by the state in which municipalities are located, their regional type, or their size.[19] Instead, within each of these categories, there is considerable leeway regarding the use of different

[17] Note that most German states feature all four regional types, and that regional types are not constrained by state boundaries. That is, a small rural municipality that lies close to a regiopolitan city in a neighboring state will be classified as "rural close to urban".

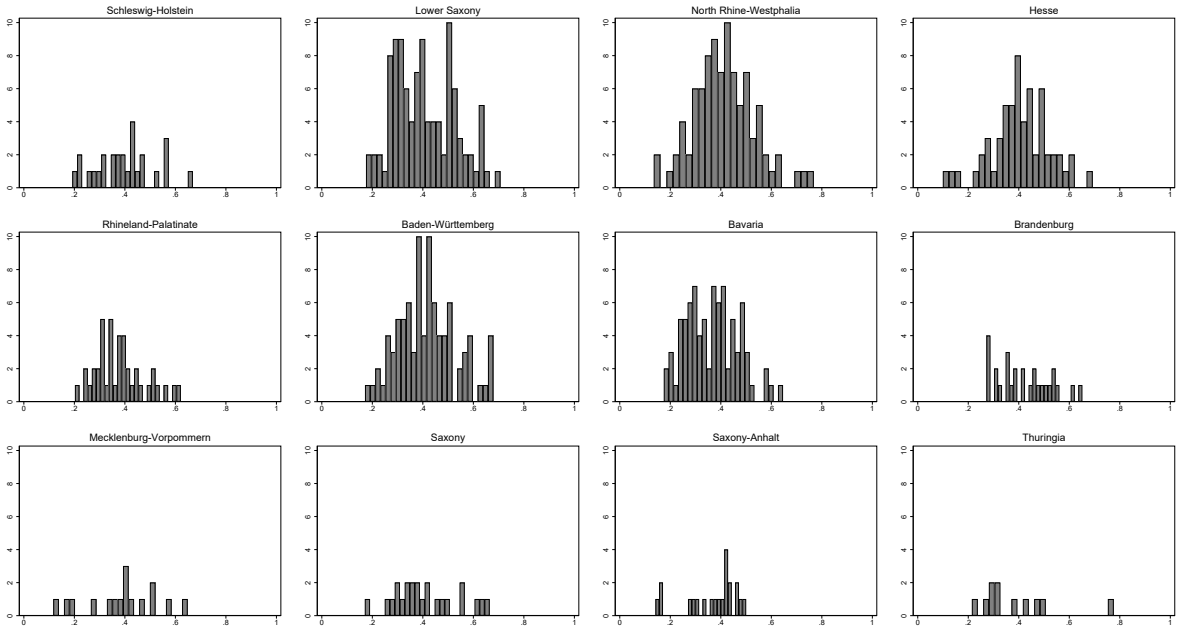
[18] According to GKBIK7, there also exists a further bracket 0 - 2,000. As explained above, municipalities in this bracket were not contacted for our survey.

[19] In fact, a Shapley value decomposition conducted in Appendix B.4 reveals that state, regional type and population size only explain 2.6, 0.9, and 11.0 percent of the management score variation, respectively. The decomposition also considers further potential sources of this variation.

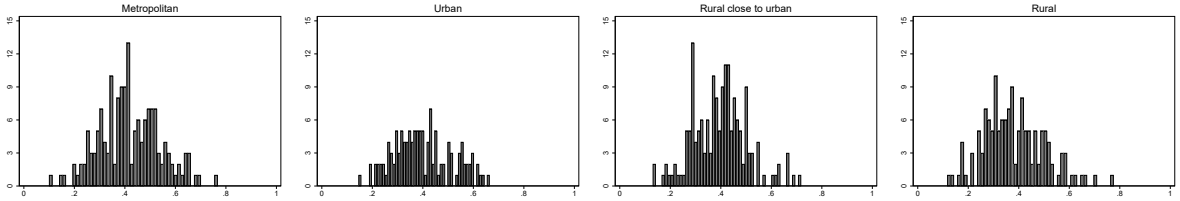
management practices, and municipalities exploit this leeway to differing degrees.

Figure 3: Frequency distribution of the management score by state, regional type and population size

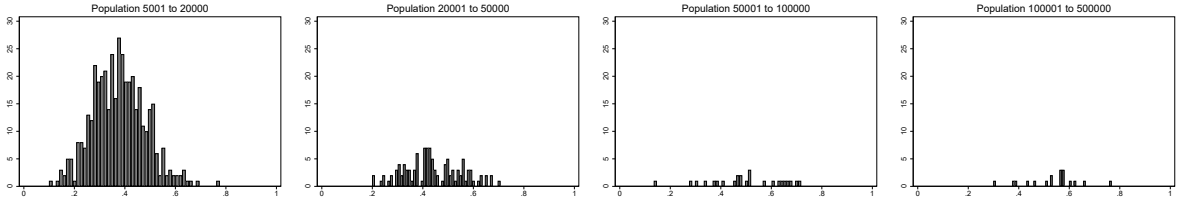
(a) By state



(b) By regional type



(c) By population size bracket



Notes: Each plot shows a management score frequency distribution (by definition, the management score lies between 0 and 1). Panel (a): Municipalities located in 13 out of the 16 German states participated in the survey. The three city states Berlin, Bremen, and Hamburg (each constituting one single municipality) did not participate. In the (small) federal state of Saarland, five municipalities participated (with management scores ranging from 0.21 to 0.5). This state is omitted in the figure. Panel (b): The population size brackets "2,001 - 5,000 inhabitants" (17 observations) and "more than 500,000 inhabitants" (3 observations) are omitted in the figure.

Finally, as a preliminary benchmarking exercise, we compare the obtained management scores of municipal administrations with those of private-sector firms in Germany. We use World Management Survey data made available by Békés and Kézdi (2021a,b), based on the 18-question survey

of Bloom and Van Reenen (2007) covering performance measurement, goal setting, operations, and human resource management policies. It contains the management scores for a total of 424 manufacturing firms in Germany, collected between 2004 and 2014 (see Appendix B.2 for details). Comparing Figure 2 with the left panel of Figure 7 (also in Appendix B.2) illustrates that management scores are on average lower for municipalities compared to manufacturing firms (mean scores of 0.4 and 0.5, respectively). Moreover, the right panel of Figure 7 shows that the cumulative management score distribution for municipalities is first-order stochastically dominated by that of firms. This tentative analysis suggests that in Germany, municipalities use structured management practices less intensively than private manufacturing firms.

5 Municipalities' objectives and performance measurement

This section addresses research question Q2. That is, we clarify the objectives of German municipalities (Section 5.1) and introduce the indicators used to measure performance relative to these objectives (Section 5.2). This is a prerequisite for the subsequent analysis of the management-performance link (research question Q3) in Section 6.

5.1 The legally stipulated objectives of municipalities

In the private sector, firms are usually assumed to maximize profits. This is a relatively well-defined objective, and direct and informative performance measures such as annual profit, sales, market share, return on equity, or total factor productivity are often readily available.[20]

For public sector organizations such as municipalities this is usually different. Instead, municipalities in Germany are supposed to follow a broader objective of providing an attractive environment for citizens and businesses, taking into account the legal and institutional setting in which they operate. For example, according to Article 57 of the municipal code of the state of Bavaria, municipalities shall "create and sustain (within the boundaries of economic feasibility) the public facilities which are necessary for the economic, social and cultural well-being of its citizens".[21]

This arguably more elusive objective function of municipalities renders performance measurement more challenging compared to the case of private firms. For example, even if detailed information on some dimension of municipal performance were available (e.g. time to reach a decision on a given type of request), such a one-dimensional performance measure might not be suitable to compare performance across municipalities. The reason is that not all municipalities might optimize along this dimension to provide an attractive environment for their stakeholders. For example, while some municipalities might prioritize good living conditions for citizens, others might focus instead on providing a good environment for businesses. As a consequence, we assemble a set of indicators of municipal performance from a variety of domains, which we consider as (more) informative about the overall degree to which municipalities fulfil their mission

[20] For a critical discussion of the goals pursued by private firms, see e.g., Tirole (2001).

[21] See https://www.gesetze-bayern.de/Content/Document/BayGO/true (only available in German).

as defined by law.

5.2 Measures of municipal performance

The indicators of municipal performance are obtained from the *Bertelsmann Stiftung*, one of the largest foundations in Germany. The Bertelsmann Stiftung engages in a variety of projects in areas such as education, health, or the economy.[22] They also provide a database with rich annual statistical information on all German municipalities with a population size of at least 5,000 inhabitants.[23] This information is compiled from a variety of sources such as statistical offices, ministries, and other agencies (both at the state and federal level). Their resulting database contains information at the municipal level from categories such as finance, demographics, education, infrastructure, and migration, and the Bertelsmann Stiftung kindly allowed us to use it for the present study. After counseling with KGSt and other expert practitioners, we consider a diverse set of eight measures from the Bertelsmann data that arguably are informative indicators of municipal performance. For all of these indicators, higher values are interpreted as better performance.

First, with respect to the municipal goal of providing an attractive environment for (new) citizens and (new) businesses, we consider the following indicators (all measured at the municipal level, monetary variables in Euro, variable names in italics):

1. *Net migration*: the number of people moving into the municipality minus the number of people moving away in a given year (per 1,000 current inhabitants). We use this as a measure of a municipality's attractiveness as a place to live for current and new citizens.
2. *Academic workforce*: the share of employees with an academic degree relative to the total employed workforce in the municipality. This variable measures the municipality's ability to attract high-skill jobs.
3. *Start-ups*: the number of newly founded firms per 1,000 inhabitants. We use this variable as a measure of a municipality's attractiveness for founders.
4. *Broadband coverage*: the share of private households with an internet broadband connection of 50 Mbit/s or larger. We use this variable as a measure for the quality of the municipality's digital infrastructure.
5. *Business tax base*: per-capita tax base (i.e., loosely speaking, corporate profits) on which municipalities levy a business tax (*Gewerbesteuer*) to be paid by establishments residing in the respective municipality. We interpret a high tax base as an indicator that a municipality provides an attractive environment for businesses to reside there and to enable them to generate profits. The business tax base is calculated by dividing a municipality's (per-capita) business tax revenue by the business tax rate (*Hebesatz*), which is set by municipalities.
6. *Income tax revenue*: the per-capita revenue from income tax paid by citizens and partnerships. In the same spirit as with the previous variable, we use this as an indicator that

[22] See https://www.bertelsmann-stiftung.de/en/home.

[23] See their internet portal "Wegweiser Kommune" at https://www.bertelsmann-stiftung.de/de/unsere-projekte/wegweiser-kommunede (available in German only). While being contained in the database, it turned out that in 17 out of the 600 municipalities participating in the management survey, the population size was less than 5,000 (see also Table 8 in Appendix B.1).

a municipality provides an attractive environment for citizens and partnerships to reside there and to generate income. Importantly, in contrast to the business tax, German municipalities have no influence on the income tax rate, which is set at the federal level. Instead, they receive as income tax revenue an amount that is by and large proportional to the income tax accruing in their constituency.[24] Therefore, income tax *revenue* is arguably a good indicator for municipalities' income tax *base*.

German municipalities are granted a considerable degree of financial autonomy (see Article 28 of the German constitution). For example, they can (within legal and fiscal limits) decide on their expenditures (e.g. the size of the workforce in the administration or which infrastructure projects to finance) and the level of debt. As a result, also fiscal health seems to be an important measure of municipal performance, and we consider two basic indicators:

7. *Primary balance*: the difference (per capita) between income (e.g. from taxes) and expenditures, net of interest payments and repayment of debts. This is a prominent indicator for the overall financial health of a fiscal entity. A larger primary balance indicates that a municipality has more leeway, for example with respect to quality and/or scope of service provision or with respect to infrastructure investments.
8. *Financial balance*: the difference (per capita) between the financial income (e.g. dividends and interest) and financial expenditures (e.g. interest payments made to creditors). This indicator measures how strongly the municipal budget is constrained by obligations from outstanding debt.

We have access to the Bertelsmann data for the period from 2006 until 2019. For our baseline analysis, we take 5-year averages over the period 2015-2019 to smooth out fluctuations.[25] Table 1 provides descriptive statistics for the above performance indicators. As can be seen, municipalities are widely dispersed with respect to all measures. Moreover, our sample of the 600 municipalities that have participated in the management survey seems to be representative in the sense that the differences between participating and non-participating municipalities are quite small. For example, as for mean comparisons, seven out of the eight corresponding t-tests are not statistically significant at the 10 percent level, and only for *Primary balance* there is a significant difference between the two means ($p = .05$). As shown in Table 2, also the pairwise correlations between our eight indicators are relatively small.

In addition to these eight performance indicators, we also calculate a composite measure, the *Municipal Performance Index* (MPI). The construction of the MPI follows the standard procedure for *z-scores*. That is, in a first step, we determine the z-score for each of the eight indicators, i.e. we normalize it to exhibit a mean of zero and a standard deviation of one. In a second step, the average of these eight z-scores yields an index value for each municipality. Finally, the MPI

[24] For a detailed discussion of the distribution of income tax revenue from the federal to the municipal level, see https://www.bundesfinanzministerium.de/Content/DE/Standardartikel/Themen/Oeffentliche_Finanzen/Foederale_Finanzbeziehungen/Kommunalfinanzen/GemeindeanteilESt-2015.pdf (available in German only).

[25] In robustness checks, we verify that our main results uphold when considering alternative time periods (see Appendix B.3).

Table 1: Indicators of municipal performance: Descriptive statistics for participating and non-participating municipalities

Indicator	Participate	N	Mean	SD	25%	50%	75%
Net migration	yes	583	7.4	7.3	3.6	6.8	9.7
	no	2,443	7.3	5.6	4.0	7.0	10.3
Academic workforce	yes	583	9.5	5.2	6.2	8.2	10.9
	no	2,444	9.7	5.5	6.4	8.5	11.4
Start-ups	yes	582	6.6	1.8	5.4	6.5	7.8
	no	2,443	6.6	2.2	5.2	6.5	7.7
Broadband coverage	yes	583	80.4	17.9	72.5	85.3	94.1
	no	2,444	79.5	20.0	71.3	86.4	94.3
Business tax base	yes	600	107.1	104.3	57.0	83.2	121.2
	no	2,483	116.5	187.5	57.1	85.3	125.9
Income tax	yes	600	462.2	134.8	368.7	460.8	544.6
	no	2,483	468.7	142.6	365.6	470.6	569.8
Primary balance	yes	600	228.1	191.9	119.0	200.0	312.6
	no	2,479	248.3	234.3	128.4	222.4	332.0
Financial balance	yes	600	-7.4	32.3	-21.3	-6.4	4.4
	no	2,479	-7.2	34.8	-21.4	-6.2	6.0

Notes: The table shows descriptive statistics for the eight indicators of municipal performance for both (i) the 600 participating municipalities for which a management score was calculated and (ii) the 2,483 non-participating municipalities with a population larger than 5,000 for which no management score could be calculated. The unit of observation for each indicator is the 5-year average from 2015 to 2019 (the five most recent years of availability) in the respective municipality. For some of the 600 participating municipalities, some indicators are not available during the five-year period. *Business tax base*, *Income tax*, *Primary balance* and *Financial balance* are measured per capita (in Euro), *Net migration* and *Start-ups* are measured by 1,000 inhabitants, *Academic workforce* and *Broadband coverage* are measured as shares. Mean comparisons (t-tests) between participating and non-participating municipalities show no significant difference for seven out of the eight indictors (exception: *Primary balance*, $p = .05$).

measure is obtained by normalizing the index values again to exhibit a mean of zero and a standard deviation of one.

Before turning to our results, it should be noted that in contrast to related studies in the private sector such as e.g. Bloom and Van Reenen (2007), Bloom, Sadun, and Van Reenen (2012, 2015c) or Bloom et al. (2019), any connection between the management practices employed in municipal administrations and the above indicators will likely be rather indirect, as such practices cannot directly affect municipal performance. For example, when certain management practices in place ensure an effective provision of municipal services, this will not directly lead to more business tax revenue, but might be one factor in a firm's decision to settle in the respective municipality (and to pay its business tax there in the future). For this reason, these indicators might be noisier measures of performance compared to those employed in studies of private sector firms. In this sense, our setting is somewhat stacking the deck against finding a (positive) statistically significant relationship between the management score and municipal performance. Nevertheless, our results

reported below strongly support the view that such a relationship in fact systematically exists.[26]

Table 2: Pairwise correlations of indicators of municipal performance

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
(1) Net migration	1.000							
(2) Academic workforce	0.112*	1.000						
(3) Start-ups	0.148*	0.206*	1.000					
(4) Broadband coverage	0.132*	0.241*	0.331*	1.000				
(5) Business tax base	0.078	0.397*	0.319*	0.186*	1.000			
(6) Income tax	0.053	0.351*	0.453*	0.333*	0.284*	1.000		
(7) Primary balance	0.170*	0.178*	0.348*	0.166*	0.698*	0.415*	1.000	
(8) Financial balance	0.009	0.155*	-0.041	0.031	0.196*	-0.105	-0.030	1.000

Notes: * indicates statistical significance at the 10 per cent level.

6 Relationship between management and performance

This section addresses the management-performance link (research question Q3), i.e. whether the observed heterogeneity in municipal management (measured by the management score) is systematically related to municipal performance. Section 6.1 presents our main results, documenting a positive and economically sizeable relationship between the management score and each of the eight municipal performance indicators as well as the management performance index (MPI). These results are robust in various alternative specification and in a bounding exercise in the spirit of Oster (2019). Section 6.2 then studies which of the six management categories covered in the survey are particularly important in this respect. We do this through a regression analysis and a Shapley value decomposition of the explained variation. Finally, in Section 6.3 rather than invoking the management score, we employ unsupervised machine learning (LDA) to identify "management styles" in the sense of characteristic patterns in the survey responses. We show that these management styles are also systematically related to municipal performance.

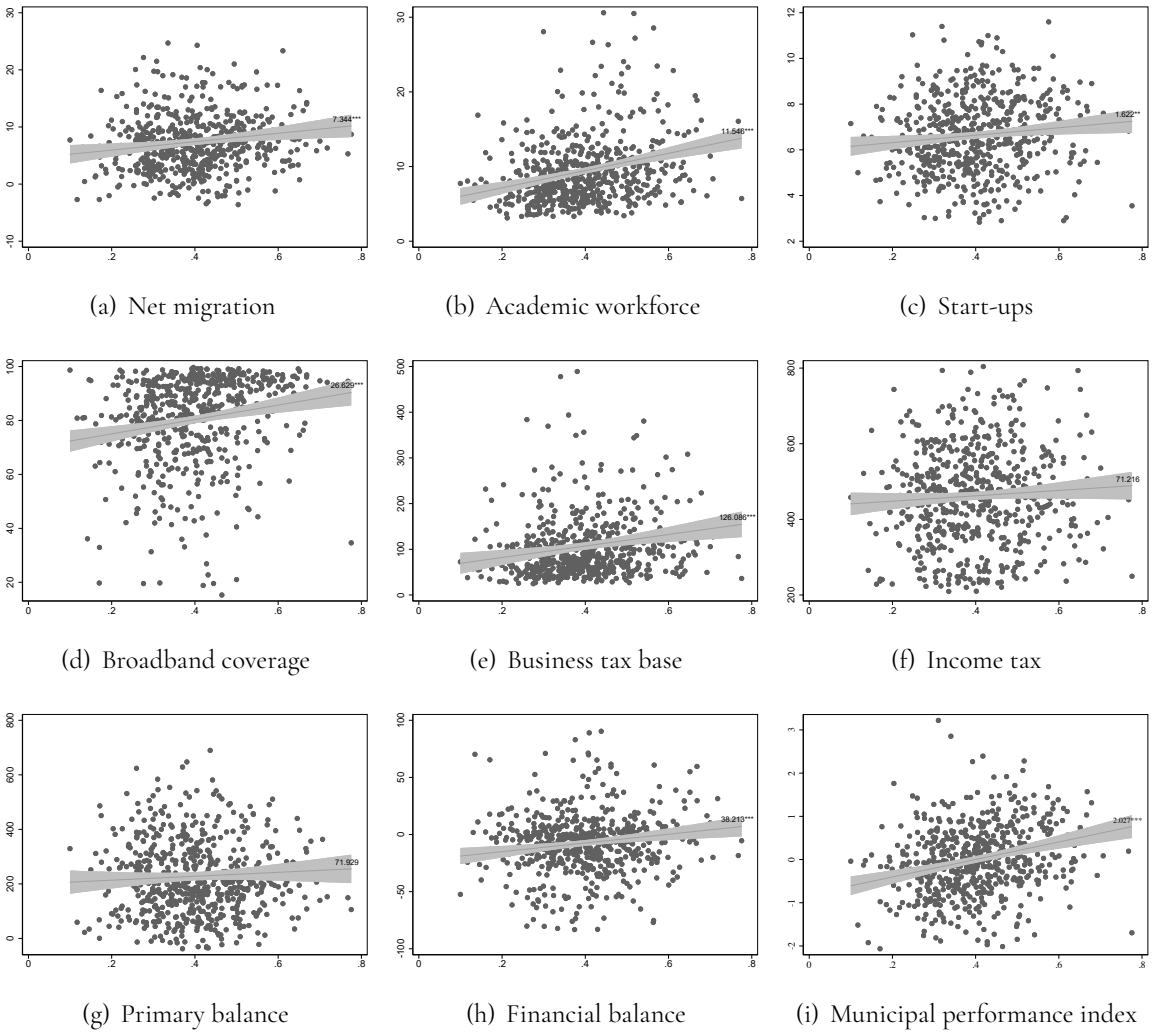
6.1 Management score and municipal performance

As a first step, we consider the raw data. Figure 4 displays scatter plots relating each indicator (on the vertical axis) to the management score (on the horizontal axis). Each panel of Figure 4 also includes a fitted regression line and the respective 95% confidence interval resulting from a simple univariate OLS regression without clustering of standard-errors. Note that according to

[26] In the preregistration, we had also envisioned to elicit customer satisfaction with municipal services. To this end, we have designed and conducted two online surveys (one among Facebook users with a focus on private citizens and one among all firms registered in any of the 600 municipalities participating in our management survey). In both of these surveys, the number of respondents was unfortunately very low (less than 1 per cent, leading to very low numbers of observations per municipality) such that the obtained data was not amenable to sound statistical analysis. We have also elicited the Google reviews of the municipal offices of all 600 participating municipalities. Again, for the vast majority of them, the number of reviews was very small and hence too noisy to be useful.

each of eight indicators and the composite MPI, municipal performance is positively related to the management score, and in seven cases this relationship is statistically significant.

Figure 4: Management score and municipal performance measures (raw data)



(a) Net migration (b) Academic workforce (c) Start-ups

(d) Broadband coverage (e) Business tax base (f) Income tax

(g) Primary balance (h) Financial balance (i) Municipal performance index

Notes: Each panel depicts the management score on the horizontal axis and the respective performance measures on the vertical axis (averages over the 5 most recent years of availability, 2015 to 2019). Each panel also show the fitted line and the 95% confidence interval resulting from a univariate OLS regression (without clustering of standard-errors) between the management score and the respective performance measures, where *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

In a second step, we perform a more thorough regression analysis with the respective indicator as the dependent variable and the management score as the key independent variable. For each indicator of municipal performance, we estimate four regression models. In all of these regressions, standard errors are clustered at the state level. Participating municipalities are located in 13 out of the 16 German states. Given the relatively low number of 13 clusters, we employ a (wild cluster) bootstrap procedure (see Cameron et al., 2008). The first (baseline) model does not include additional controls. In the second model, we use state dummies to control for state-specific effects (e.g. legal, political, and economic factors) under which the respective municipalities located in a given state operate. In the third model, we additionally include dummies that control for the

regional type of each municipality, using the *RegioStaR* classification as explained in Section 4.2. Although all of our performance indicators are defined on a per capita basis or as shares, it might still be the case that larger municipalities differ qualitatively from smaller ones. Consequently, in the fourth model, in addition to state and regional type dummies we control for municipalities' size, using the six population size brackets as defined in Section 4.2. Note that while municipalities' exact population size varies over time (and might depend on how well a given municipality is managed), the population size bracket does not vary across the sample period 2015-2019 for 99% of the 600 participating municipalities.[27]

The results of all regressions are shown in Table 3, which is structured as follows: Each row corresponds to a given performance indicator as the dependent variable, and it shows the coefficient and p-value of the management score in each of the four regression specifications discussed above (while the coefficients of control variables are not displayed). For example, the entry 7.34 (5.92) in the upper left (right) cell is the coefficient of the management score for the effect on net migration when not adding any further controls (when controlling for state, regional type, and population size bracket).

The regression analysis largely confirms the findings of Figure 4. In particular, all 36 coefficients for the management score indicate a positive relationship with performance. Moreover, 25 of them are statistically significant, and in the other cases the p-values are generally not far off. With respect to column (4), note that there is a positive and significant correlation between population size and the management score (r = 0.27), which might pick up part of the effect of the latter. For the composite MPI, the coefficient of the management score is significant in all four specifications. Our findings thus clearly point to a systematic positive relationship between the degree of structured management in municipal administrations (as measured by the management score) and municipal performance.

Robustness As shown in Appendix B.3, the main results as reported in Table 3 are qualitatively robust (i) when averaging the outcome measures over the three most recent years (instead of five years) and when looking at the most recent year in the data only (see Tables 9 and 10, respectively), and (ii) when using the z-score of the survey answers instead of the management score (see Table 11). The z-score is an alternative measure of structured management employed in the WMS literature (see e.g., Scur et al., 2021), and in our sample its correlation with the management score is 0.98.

Finally, performing the bounding exercise suggested by Oster (2019), our results are robust with respect to potential omitted variable bias. This approach uses the changes of the management score coefficient and R^2 between the baseline model and the richest model (see columns (1) and (4) of Table 3) to estimate the degree of proportional selection on unobservables relative to observables that would be required to reduce the coefficient to zero (i.e. δ). We find that such

[27] Seven out of these 600 municipalities are assigned to a different population size bracket in one out of the five years of the sample period. To these municipalities we assign the 2015 population bracket, which in each of these cases was also the modal value. The results of column (4) of Table 3 remain robust when dropping these seven municipalities from the sample.

Table 3: Management score and municipal performance

	(1)	(2)	(3)	(4)
Net migration	7.34***	7.27***	6.68***	5.92***
	(0.00)	(0.00)	(0.00)	(0.00)
Academic workforce	11.55***	11.46***	9.93***	4.16
	(0.00)	(0.00)	(0.00)	(0.18)
Start-ups	1.62***	1.24***	0.96**	0.49
	(0.00)	(0.00)	(0.01)	(0.25)
Broadband coverage	26.63***	27.41***	24.48***	11.34
	(0.00)	(0.00)	(0.00)	(0.13)
Business tax base	126.09***	144.66***	140.35**	117.79
	(0.01)	(0.01)	(0.03)	(0.14)
Income tax revenue	71.22	80.71***	37.32	52.24**
	(0.17)	(0.00)	(0.14)	(0.02)
Primary balance	71.93	126.36	124.98	127.79
	(0.27)	(0.12)	(0.12)	(0.20)
Financial balance	38.21**	36.50**	36.12***	20.95
	(0.03)	(0.04)	(0.00)	(0.14)
Municipal performance index (MPI)	2.03***	2.09***	1.85***	1.22**
	(0.00)	(0.00)	(0.00)	(0.04)
State dummies	No	Yes	Yes	Yes
Regional type dummies	No	No	Yes	Yes
Population bracket dummies	No	No	No	Yes

Notes: Each row corresponds to a given performance indicator as dependent variable (five-year averages), and shows the coefficient and p-value of the management score in each of the four regression models discussed in the main text (while the coefficients of control variables are not displayed). For example, the entry 7.34 (5.92) in the upper left (right) cell is the coefficient of the management score for the effect on net migration when not adding any further controls (when controlling for state, regional type, and population bracket). In all regressions, standard errors are clustered at the state level (employing a wild cluster bootstrap), where *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively. The number of observations is 600, except for *Net migration*, *Academic workforce* and *Broadband coverage* where it is 583, and for *Start-ups* where it is 582 (see Table 1). For the missing municipalities, the respective indicators are not available.

selection would have to be large (see Table 12). In particular, for all performance indicators we obtain (absolute) δ-values above the critical value of 1 as suggested in Oster (2019).

Economic Significance In order to gauge the economic significance of our findings, we perform an interquartile comparison between municipalities with a management score at the 25%- and

Table 4: Economic significance

	(1)	(2)	(3)	(4)
Indicator of municipal performance	Mean	SD	Coefficient of management score	Effect of interquartile score change of 0.17
Net migration	7.4	7.3	5.92	1.01 (14%, 14%)
Academic workforce	9.5	5.2	4.16	0.71 (7%, 14%)
Start-ups	6.6	1.8	.49	0.08 (1%, 5%)
Broadband coverage	80.4	17.9	11.34	1.93 (2%, 11%)
Business tax base	107.1	104.3	117.79	20.02 (19%, 19%)
Income tax	462.2	134.8	52.24	8.88 (2%, 7%)
Primary balance	228.1	191.9	127.79	21.72 (10%, 11%)
Financial balance	-7.4	32.3	20.95	3.56 (-48%, 11%)

Notes: Columns 1 and 2 display the mean and standard deviation of each of the eight indicators of performance of the participating municipalities (see also Table 1). Column 3 re-states the coefficient for the management score from the richest regression specification (4) in Table 3. Column 4 shows the absolute ceteris-paribus change in the respective performance measure for the interquartile difference in the management score (with values of 0.31 and 0.48 at the 25% and 75% percentile, respectively, leading to a difference of 0.17). That is, the entries in column 4 are obtained by multiplying the respective coefficient in column 3 by 0.17. The first (second) number in brackets denotes the absolute change as a fraction of the respective mean (standard deviation). For example, for the case of net migration, we have: $1.01 = 0.17 \cdot 5.92$ ($14\% = 1.01 \cdot 100/7.4$, $14\% = 1.01 \cdot 100/7.3$).

the 75%-percentile, where the respective score values are .31 and .48, leading to a difference of .17. In Table 4, for each performance measure we then evaluate this score difference of .17 with the respective coefficient of the management score in the richest regression specification as reported in Table 3, column (4). That is, the first entry in column (4) of Table 4 denotes the absolute change of the respective performance measure and is obtained by multiplying the respective coefficient in column (3) by the score difference of .17. The two entries in brackets state this absolute change as a fraction of the mean and the standard deviation, respectively, for the performance indicator under consideration. For example, for the case of net migration, the absolute change is $1.01 = .17 \cdot 5.92$, which constitutes 14% of both the mean (7.4) and the standard deviation (7.3). Overall, the economic effects appear to be sizeable.

6.2 Management subscores and municipal performance

Recall from Section 3.1 that our survey contains questions from six pre-specified management categories (goal setting, monitoring of service quality, organizational learning, structured processes, human resource management, and digitization). In this section, we analyze the respective effects of these categories in more detail.[28] In a first step, we construct a *management subscore* for each category from the respective survey questions.[29] We then regress municipal performance (mea-

[28]This part of the analysis was not addressed in the preregistration. We have nevertheless included it as it turns out to be helpful for the subsequent (preregistered) analysis of municipal management styles.

[29]The subscores are contructed in the same way as the overall management score used so far. See Appendix A for more information on which questions enter which subscore.

sured by the composite Municipal Performance Index (MPI) introduced in Section 5) on these subscores. The results are shown in Table 5. There, we consider the same four regression models as in main Table 3 above, but in Table 5 each column refers to just one regression, i.e. each coefficient indicates the effect of the respective subscore on the MPI. As can be seen, it is particularly the use of management practices related to the monitoring of service quality and human resource management that exhibit a statistically significant relationship with municipal performance.

Furthermore, we assess each management subscore's relative contribution to overall explanatory power. That is, for each of the four specifications of Table 5, we conduct a Shapley value decomposition of the explained variation (R^2).[30] The Shapley values represent the average marginal increase in model fit attributable to each of the six management subscores. We find that across all specifications, the subscore for human resource management has by far the largest effect, followed by monitoring of service quality, and organizational learning, roughly by factors of three and six, respectively. For example, for the specification shown in the rightmost column of Table 5, the Shapley values for these three subscores are 11 percent, 4 percent, 2 percent, respectively, while the remaining three Shapley values sum up to less than 2 percent.

The topicality of human resource management policies is corroborated by the answers to the three free-form questions included in the municipal survey (questions Q47-Q49). We asked respondents what they perceive as major challenges for implementing high-quality municipal services, and it turns out that topics related to human resource management, such as flexible remuneration, ensuring an attractive workplace, or attracting talent loom large in the answers. Taken together, this suggests that in discussions of changing management practices in the public sector, human resource management should receive particular attention.

The analysis of subscores provides a first indication on the relative importance of the various management categories. In a next subsection, we analyze whether management practices are systematically used in combination – thereby giving rise to distinct municipal *management styles*– and whether there exists a systematic relationship between management styles and municipal performance.

6.3 Management styles and municipal performance

In this section, we investigate whether there are distinct patterns in the answers to the survey questions across municipalities, in particular, whether certain sets of practices are adopted together. Thereby, our aim is to gain a better understanding of different "management styles" employed in these public sector organizations. Hence, this analysis complements the evidence provided in Section 4.1 on the anatomy of management practices in local government.

To analyze the high-dimensional survey data, we apply unsupervised machine learning in the form of *Latent Dirichlet Allocation* (LDA, see Blei et al. 2003) to classify municipalities according to their management styles. Compared to other algorithms (such as principal component analysis

[30] Shapley's seminal 1953 contribution (reprinted in Shapley, 1988) is rooted in cooperative game theory and proposes a solution for bargaining games with more than two players. More recently, a growing body of research has employed the underlying idea for decomposing statistical measures such as R^2 (see e.g. Israeli, 2007; Huettner and Sunder, 2012; Biasi, Lafortune, and Schönholzer, 2025).

Table 5: Management subscores and municipal performance

	(1)	(2)	(3)	(4)
goal setting	-0.17	-0.03	-0.08	-0.22***
	(0.17)	(0.66)	(0.23)	(0.01)
monitoring of service quality	0.57*	0.69**	0.63**	0.43
	(0.07)	(0.03)	(0.01)	(0.18)
organizational learning	0.27	0.41	0.42	0.46
	(0.45)	(0.20)	(0.17)	(0.13)
structured processes	-0.17	0.23	0.10	-0.05
	(0.88)	(0.13)	(0.37)	(0.73)
human resource management	1.95***	0.82*	0.74**	0.72**
	(0.01)	(0.06)	(0.01)	(0.01)
digitization	0.12	0.06	0.13	0.09
	(0.60)	(0.72)	(0.51)	(0.64)
R^2	0.08	0.32	0.40	0.44
Adj. R^2	0.07	0.30	0.37	0.41
State dummies	No	Yes	Yes	Yes
Regional type dummies	No	No	Yes	Yes
Population bracket dummies	No	No	No	Yes

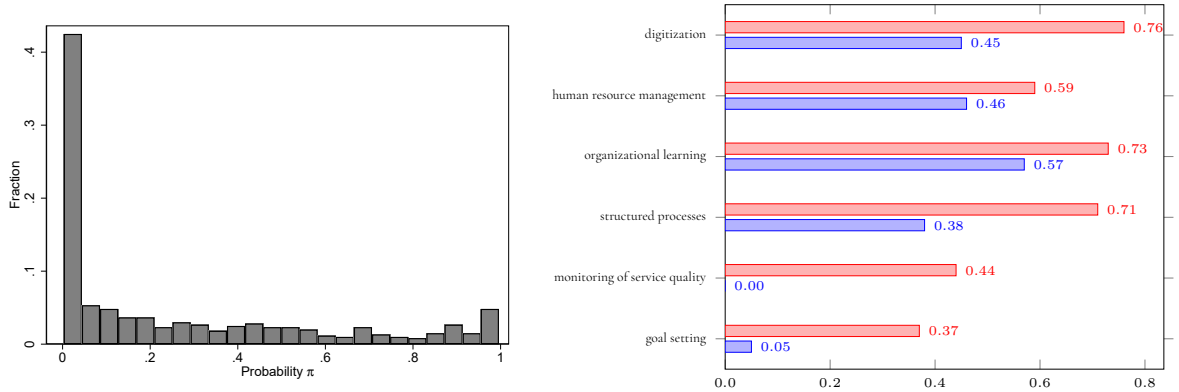
Notes: The table reports regressions where the dependent variable is the municipal performance index (MPI), i.e., the z-score of the eight performance indicators. The key independent variables are the management subscores for the six management categories in the survey. Specification (1) corresponds to a baseline with no additional controls, while specifications (2) – (4) consecutively include dummies for state, regional type, and population size (as explained in Section 6.1). Each entry in the table shows the coefficient and the p-value for the respective management subscore. In all regressions, standard errors are clustered at the state level (employing a wild cluster bootstrap), where *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

or k-means analysis), the LDA allows to include a large number of variables in the classification and also to take into account correlations between combinations of the variables. The LDA is also a suitable tool for the analysis of survey data (see e.g., Airoldi et al., 2014). In the context of economics, it has been employed to analyze central bank protocols (Hansen et al., 2018), CEO management styles (Bandiera et al., 2020), and management practices in Spanish manufacturing firms (Englmaier et al., 2025).

Our approach follows Bandiera et al. (2020). That is, we postulate two "pure" management styles, which the LDA then extracts from the survey responses. Each municipality's management style is then expressed as a convex combination of these two pure styles. In particular, to each municipality the LDA assigns probabilities π and $1 - \pi$ as the probability weights on either pure style. Panel (a) of Figure 5 shows the histogram of the probability π across municipalities. As can be seen, roughly 40 percent and 5 percent of municipalities closely resemble one of the pure styles, respectively. All other municipalities are (more or less uniformly) distributed between these extremes.

In order to interpret the management styles identified by the LDA, we investigate key management features of the municipal administrations whose management style either exactly or very closely matches one of the two pure styles (where we use $\pi \leq 0.1$ and $\pi \geq 0.9$ as cutoffs, and refer to these municipalities as "Type 0" and "Type 1", respectively). Panel (b) of Figure 5 displays the differences between these two styles for each of the six management subscores (see Section 6.2). As can be seen, municipalities of Type 1 exhibit substantially higher scores in all six categories compared to their Type 0 counterparts. Hence, the analysis suggests that the municipalities in our sample do not specialize in different management categories. Rather it appears that they either actively manage their affairs, or merely passively administer them.

Figure 5: Management styles using the LDA



(a) Relative frequencies of probability π

(b) Subscores of Type 1 (red) and Type 0 (blue) municipalities

The difference between the two types is particularly pronounced for the categories *monitoring of service quality* and *goal setting*, where municipalities of Type 0 are hardly active (if at all), as indicated by management subscores that are (close to) zero. In the light of these differences, one might refer to municipalities of Type 1 and 0 as "actively managed" and "administered", respectively. Strikingly, the latter constitute half (49.8%) of all municipalities in our sample. Moreover,

78% of municipalities are closer to the "administered" style than to the "actively managed" style (i.e. they are characterized by $\pi \leq 0.5$).

Using the LDA results, we have also investigated the relationship between each municipality's management style and municipal performance. In particular, we use the probability weight π on the "actively managed" Type 1 as an alternative measure of structured management (instead of the management score used in main analysis). It turns out that π and the management score are positively correlated (pairwise correlation: 0.70, significant at the 1 percent level). Moreover, Table 13 in Appendix B.3 replicates the analysis of main Table 3 using π instead of the management score, and the results are robust, i.e. actively managed municipalities perform better than administered ones.

7 Municipal performance in a specific task: Provision of online services

As discussed in Section 5, German municipalities are legally required to promote the well-being of their citizens. In Section 6, we evaluated performance with respect to this rather elusive objective by considering performance indicators from a variety of domains that might capture "well-being" in a broad sense. We find a systematic positive relationship between the management score and these measures of municipal performance (see Table 3). In this section, we investigate whether a positive management-performance link also exists with respect to to a *specific* service, that is more directly controlled by municipalities.

Due to the German federal structure and a lack of coherent reporting requirements, such task-specific performance data is generally unavailable at the municipal level. We exploit an exception to this, the number of services provided digitally ("online services") by each municipality. In April 2024, this information has become publicly available through the "Online Access Act" (*Online-Zugangsgesetz*, OZG) of 2017, and of a revised version in 2024. The OZG is a federal initiative to boost the online provision of public services. It was motivated by the fact that Germany is lagging behind in digitizing its public administrations. For example, in a EU-wide comparison based on the *Bitkom-DESI-Index 2025*, it ranks only 21st out of the 27 EU member states.[31]

The implementation of the OZG has lead to a maximum of 423 online services that municipalities may provide.[32] We have split this total number into online services to citizens (225) and businesses (198), respectively. Importantly, municipalities have discretion over *when* to start providing each individual service online, and they report the provided services to a federal registry. These registry data are now publicly available from a dashboard operated by the federal government.[33] When accessing the dashboard, it delivers a snapshot of the online services provided by

[31] See https://www.bitkom.org/Presse/Presseinformation/Digitalisierung-Deutschland-EU-Vergleich-Platz-14, available in German only. *Bitkom* is Germany's leading digital industry association, representing more than 2,000 companies from the IT, telecommunications, and digital sectors. See also the *EU Digital Decade 2025* country report for Germany (EU Commission, 2025, pp. 17).

[32] In addition, there are further services to be provided by the states or the federal administration.

[33] See https://dashboard.digitale-verwaltung.de/ (in German only).

municipalities at the respective date. We use the dashboard data from the first date of availability (April 2024).

Table 6: Municipal online services

	(1)	(2)	(3)	(4)	(5)	(6)
	To citizens	Participate		To businesses	Participate	
#		yes	no		yes	no
1	Dog registration (+)	49 (1)	50 (2)	Business registration/licensing (+)	72 (1)	70 (1)
2	Birth certification (+)	49 (2)	49 (3)	Permits weekly/special markets	47 (3)	50 (2)
3	Residence certification (+)	47 (3)	50 (1)	Special road use/traffic restrictions	48 (2)	49 (3)
4	Change of address (+)	45 (6)	47 (4)	Expert appointment/accreditation	39 (6)	45 (4)
5	Marriage certification (+)	46 (5)	47 (5)	Craft activity notification/permit (+)	43 (4)	40 (5)
6	Death certification (+)	46 (4)	46 (6)	Licensing regulated professions (+)	42 (5)	40 (6)
7	Housing benefit (+)	45 (7)	43 (7)	Recognition foreign qualifications (+)	37 (7)	37 (7)
8	Data disclosure restrictions	41 (8)	41 (8)	Entry in crafts register (+)	34 (9)	37 (8)
9	Identity card application	38 (10)	39 (9)	Building permit	34 (8)	36 (9)
10	Fireworks permit	39 (9)	39 (10)	Exemptions from closing hours	28 (12)	32 (10)

(a): Top-ten rankings

Indicator	Participate	N	Mean	SD	25%	50%	75%
Online services all	yes	600	27.8	29.2	5.0	20.0	38.0
	no	2,483	30.5	30.5	5.0	21.0	45.0
Online services citizens	yes	600	14.6	15.3	3.0	10.0	22.0
	no	2,483	15.7	15.0	3.0	13.0	25.0
Online services businesses	yes	600	13.1	15.6	1.5	8.0	17.0
	no	2,483	14.9	17.3	1.0	9.0	22.0

(b): Descriptive statistics

Notes: In panel (a), column 1 lists the top-ten online services to citizens among all 3,083 surveyed municipalities. In column 2, the first number indicates the percentage of the 600 participating municipalities providing the respective online service, and the second number (in parentheses) indicates its frequency rank among all online services to citizens provided by participating municipalities. Column 3 provides the same information for all 2,483 non-participating municipalities. Columns 4–6 proceed analogously for services provided to businesses. The symbol (+) indicates a large positive difference in the provision of the respective service between municipalities with a management score above and below the median (see the discussion at the end of Section 7). Panel (b) displays descriptive statistics for the three indicators of municipal performance with respect to online service provision. The mean differences between participating and non-participating municipalities are statistically significant at the 5-percent level (t-tests) for *Online services all* and *Online services business*.

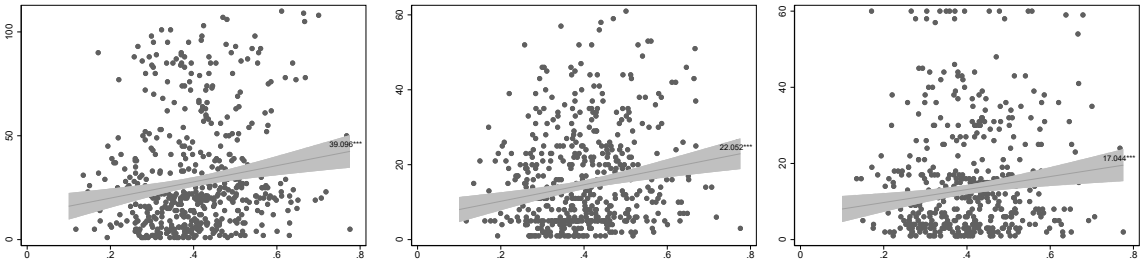
To give an idea of the services provided digitally, Table 6 (a) shows the ten most frequent online services for citizens and businesses, respectively (separately for municipalities that participated in our management survey and those that did not). For example, among all 3,083 municipalities, the registration of dog ownership is the most frequent online service to citizens (# 1). It ranks first among participating municipalities (where 49 percent provide this online service), and second among non-participating ones (where 50 percent provide it). Overall, the service provision patterns for participating and non-participating municipalities are very similar. Seven out of the services listed in Table 6 (a) are also among the 16 so-called "focus services" (*Fokusleistungen*), for

which the OZG deems swift online provision (by federal, state, or municipal administrations) particularly important due to the high demand for these services.[34]

In the following, we measure municipal performance with respect to online service provision by the number of online services provided by the respective municipality in April 2024, where we distinguish between services to citizens (*Online services citizens*), to businesses (*Online services businesses*), and the sum of the two (*Online services all*). Table 6 (b) displays descriptive statistics for these three performance indicators, again separately for participating and non-participating municipalities. The average total number of online services is approximately 28, and hence less than 10 percent of a total of 423 of potential online services. Non-participating municipalities provide slightly more online services than participating ones (where the mean comparisons for *Online services all* and *Online services business*, respectively, are statistically significant at the 5-percent level, t-tests).

We now show that better managed municipalities (again measured by their management score) are also more successful in digital service provision. Analogous to the main analysis in Section 6, we first look at the raw data. The scatter plots in Figure 6 clearly show that for each of the three performance indicators, there exists a strong and statistically significant positive relationship with the management score.

Figure 6: Management score and municipal performance in online service provision (raw data)



(a) Online services all (b) Online services citizens (c) Online services businesses

Notes: Each panel depicts the management score on the horizontal axis and the respective OZG performance measure on the vertical axis. Each panel also show the fitted line and the 95% confidence interval resulting from a univariate OLS regression (without clustering of standard-errors) of the respective performance measure on the management score, where *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

This finding is supported by a regression analysis. As Table 3 above, Table 7 shows the coefficients of the management score for each of a total of 12 regressions (i.e. four regressions for each of the three performance indicators). It can be seen that all 12 coefficients are positive and statistically significant, virtually all of them at a high level. Thereby, the coefficients of the management

[34]These services are change of address, marriage certification, housing benefits, identity card application, business registration/licensing, entry in crafts register, building permits, see e.g. Federal Ministry of the Interior (2025) and https://www.digitale-verwaltung.de/Webs/DV/DE/onlinezugangsgesetz/das-gesetz/ozg-aenderungsgesetz/fokusleistungen/fokusleistungen.html (both in German only).

score are considerably larger for services provided to citizens compared to businesses.[35]

Table 7: Management score and municipal performance in online service provision

Online services all	39.10***	29.01***	27.77***	13.53***
	(0.00)	(0.00)	(0.00)	(0.00)
Online services citizens	22.05***	19.61***	18.65***	9.67***
	(0.00)	(0.00)	(0.00)	(0.00)
Online services businesses	17.04***	9.40***	9.12***	3.86*
	(0.00)	(0.00)	(0.00)	(0.05)
State dummies	No	Yes	Yes	Yes
Regional type dummies	No	No	Yes	Yes
Population bracket dummies	No	No	No	Yes

Notes: In analogy to Table 3, each row corresponds to one of the three performance indicators for online service provision, and shows the coefficient and p-value of the management score in each of the four regression models (while the coefficients of control variables are not displayed). For example, the entry in the upper right cell is the coefficient of the management score for the effect on *online services all* when controlling for state, regional type, and population size bracket. In all regressions, N = 600, and standard errors are clustered at the state level (employing a wild cluster bootstrap), where *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

An interquartile comparison (i.e. a management score difference of .17 as above) reveals that also effect sizes are large. For example, municipalities at the 75% percentile of the management score distribution provide $.17 \cdot 13.53 = 2.30$ more online services (fourth regression model) compared to those at the 25% percentile. This corresponds to 8.27 (7.88) percent of the mean (standard deviation) of all participating municipalities. As can also be seen from Table 7, the effect is stronger for services provided to citizens compared to businesses. All in all, the results on the provision of online services complement the key finding from the main analysis in Section 6: There exists a strong and positive management-performance link in German municipalities.

In a final step, we investigate in more detail *which* online services are provided more frequently by municipalities that have a higher management score. We proceed as follows. First, we perform a median split with respect to the management score to divide the participating 600 municipalities into two groups of equal size. Second, within each group, we calculate the average provision rate for each of the 423 online services considered. Third, for each service, we calculate the between-group difference of provision rates. Hence, a positive difference indicates that the respective service is more frequently provided by municipalities with an above-median management score.[36] Finally, services are ranked by the size of between-group differences. We find (positive) differ-

[35] As in the main analysis (see Section 6.1), as a robustness check we have conducted the bounding exercises of Oster (2019). The δ values for the performance measures are 1.38, 1.86, and 1.45, respectively, and hence above the suggested threshold of one. Again, this speaks against the importance of omitted variable bias. Moreover, a Shapley value decomposition of all six management subscores reveals that, in all specifications, the digitization subscore ranks either highest or second-highest, with a share of 20 to 40 percent of the explanatory power of the six subscores.

[36] For example, suppose a specific service is provided by 75 out of the 300 municipalities with a management score above the median, and by 60 out of the 300 municipalities with a management score below the median, resulting in within-group average provision rates of 25 and 20 percent, respectively. This would result in a between-group difference of 5 for this service.

ences in the provision rate of up to 14 percentage points, and even at rank 20, the difference is still a sizeable 7.4 percentage points.[37] Many of the services with a large provision rate difference relate to municipalities' "bread-and-butter business" in the following sense: The first seven out of the top-ten citizen services shown in Table 6 (a) also appear in the top-ten of the difference ranking (indicated by (+)). Thereby, the largest difference occurs for housing benefits (14 percentage points). A similar picture emerges for business services where five out of the top-ten services also appear in the top-ten difference ranking (with the highest difference for licensing for regulated professions). Overall, in line with our previous finding of a stronger relationship between the management score and service provision for citizens (see Table 7), 14 out of the top 20 services in the difference ranking relate to citizens, and only 6 to businesses.

To summarize, evaluating the online service provision under the OZG provides task-specific evidence that aligns with our previous findings: municipalities with a higher management score also are more effective in managing the digital transformation.

8 Conclusion

This paper studies the management and performance of local government institutions in a G7 country. Adapting the World Management Survey (WMS) framework to this context, we have conducted a survey among the leaders of municipalities in Germany, eliciting the relevance of various management practices in their administrations along six management categories. The survey yields a representative sample of 600 participating municipalities, and provides an anatomy of management practices used in German municipal administrations. We show that, despite operating in a heavily regulated environment, municipalities retain considerable leeway in their choice of management practices, leading to substantial heterogeneity. Importantly, this heterogeneity persists within federal states, regional types, and population size brackets. Using tools from unsupervised machine learning (LDA) further reveals that while "actively managed" municipalities utilize structured management across all categories, the majority follow a passive, "administered" style.

We then investigate municipal performance relative to the legally stipulated, yet elusive, objective of fostering citizens' well-being. Utilizing a broad set of performance indicators across many relevant dimensions, we show that there exists a systematic, positive, and economically significant relationship between a municipality's management score and its performance. A subscore analysis shows that HR management and the monitoring of service quality are the main drivers of this association. Furthermore, the paper is the first to identify and use a Germany-wide task-specific municipal performance measure, the digital provision of public services under the Online Access Act (OZG). We also find a strongly positive management-performance link for this specific task.

Our study can inform current discussions on how to maintain and/or improve state capacity. In many countries (including Germany), public-sector administrations are under severe strain

[37] There are considerably fewer services where the difference is negative, and negative differences are also rather small in absolute terms, e.g. 4 percentage points at rank 1 (i.e. the largest negative difference) and 1 percentage point at rank 20.

due to multiple transformations that lie ahead of them (e.g. digitization, adaption to climate change, improving resilience), while demographics are often unfavorable and public finances are strained. For example, in Germany the public sector is projected to face a vacancy gap of more than 1 million employees by 2030 (see e.g., Normenkontrollrat, 2024, pp. 15-16), which is very unlikely to be closed. As a result, (potentially shrinking) resources must be used more effectively. If the established relationship between the management of municipalities and their performance is indeed causal, then our analysis would point to directions how this might be accomplished without relying on unrealistic workforce expansion, for example by adopting an active managerial mindset and employing management practices that are more structured. Moreover, our finding that many leaders of local government institutions do have managerial leeway suggests that eventual reforms should focus less on changing top-down legal frameworks and more on encouraging these local leaders to actively utilize their existing discretion. Last, but not least, the success of national digital initiatives (such as the OZG) might not only depend on the IT infrastructure and other resources provided, but also on local managerial choices. All in all, we strongly believe that how public sector organizations are managed can play an important role in improving their fitness for the challenges ahead.

References

Acemoglu, D. and J. Robinson (2012): *Why Nations Fail: Origins of Power, Poverty and Prosperity*, Crown Publishing Group.

Adelman, M., J. Baron, and R. Lemos (2020): "Managing shocks in education: Evidence from hurricane Matthew in Haiti," *World Bank, mimeo*.

Airoldi, E. M., D. M. Blei, E. A. Erosheva, and S. E. Fienberg (2014): "Introduction to mixed membership models and methods." *Handbook of Mixed Membership Models and Their Applications*, 100, 3–14.

Ashraf, N., O. Bandiera, E. Davenport, and S. Lee (2020): "Losing prosociality in the quest for talent? Sorting, selection, and productivity in the delivery of public services," *American Economic Review*, 110, 1355–1394.

Bandiera, O., A. Prat, S. Hansen, and R. Sadun (2020): "CEO behavior and firm performance," *Journal of Political Economy*, 128, 1325–1369.

Bandiera, O., A. Prat, and T. Valletti (2009): "Active and passive waste in government spending: evidence from a policy experiment," *American Economic Review*, 99, 1278–1308.

Békés, G. and G. Kézdi (2021a): *Data Analysis for Business, Economics, and Policy*, Cambridge University Press, Cambridge.

——— (2021b): *Datasets for Békés, G. and G. Kézdi "Data Analysis for Business, Economics, and Policy"*, Cambridge University Press, Cambridge.

Besley, T., R. Burgess, A. Khan, and G. Xu (2022): "Bureaucracy and development," *Annual Review of Economics*, 14, 397–424.

Biasi, B., J. Lafortune, and D. Schönholzer (2025): "What works and for whom? Effectiveness and efficiency of school capital investments across the US," *The Quarterly Journal of Economics*, 140, 2329–2379.

Blei, D. M., A. Y. Ng, and M. I. Jordan (2003): "Latent Dirichlet allocation," *Journal of Machine Learning Research*, 3, 993–1022.

Bloom, N., E. Brynjolfsson, L. Foster, R. Jarmin, M. Patnaik, I. Saporta-Eksten, and J. Van Reenen (2019): "What drives differences in management practices?" *American Economic Review*, 109, 1648–83.

Bloom, N., R. Lemos, R. Sadun, and J. Van Reenen (2015a): "Does management matter in schools?" *Economic Journal*, 125, 647–674.

——— (2020): "Healthy business? Managerial education and management in health care," *Review of Economics and Statistics*, 102, 506–517.

Bloom, N., C. Propper, S. Seiler, and J. Van Reenen (2015b): "The impact of competition on management quality: evidence from public hospitals," *Review of Economic Studies*, 82, 457–489.

Bloom, N., R. Sadun, and J. Van Reenen (2012): "Americans do IT better: US multinationals and the productivity miracle," *American Economic Review*, 102, 167–201.

——— (2015c): "Do private equity owned firms have better management practices?" *American Economic Review*, 105, 442–46.

Bloom, N. and J. Van Reenen (2007): "Measuring and explaining management practices across firms and countries," *Quarterly Journal of Economics*, 122, 1351–1408.

Buffington, C., L. Foster, R. Jarmin, and S. Ohlmacher (2017): "The management and organizational practices survey (MOPS): An overview," *Journal of Economic and Social Measurement*, 42, 1–26.

Cameron, C., J. Gelbach, and D. Miller (2008): "Bootstrap-based improvements for inference with clustered errors," *The Review of Economics and Statistics*, 90, 414–427.

Carreri, M. (2021): "Can good politicians compensate for bad institutions? Evidence from an original survey of Italian mayors," *The Journal of Politics*, 83, 1229–1245.

Carreri, M. and J. Payson (2021): "What makes a good local leader? Evidence from US mayors and city managers," *Journal of Political Institutions and Political Economy*, 2, 199–225.

——— (2024): "Local leaders and the pursuit of growth in US cities: The role of managerial skill," *Political Science Research and Methods*, 12, 113–129.

Decarolis, F., L. M. Giuffrida, E. Iossa, V. Mollisi, and G. Spagnolo (2020): "Bureaucratic competence and procurement outcomes," *Journal of Law, Economics, and Organization*, 36, 537–597.

Destatis (2021): *GV-ISys: Verzeichnis der Regional- und Gebietseinheiten Definitionen und Beschreibungen*, Available at: https://www.destatis.de/DE/Themen/Laender-Regionen/Regionales/Gemeindeverzeichnis/Administrativ/beschreibung-gebietseinheiten.pdf?__blob=publicationFile.

Deutscher Beamtenbund (2022): "Handlungsfähigkeit des Staates: Kommunen im Fokus," dbb aktuell 32/2022, dbb beamtenbund und tarifunion, Berlin.

Duflo, E., R. Hanna, and S. Ryan (2012): "Incentives work: Getting teachers to come to school," *American Economic Review*, 102, 1241–78.

Dwenger, N. and A. Gumpert (2023): "Building fiscal capacity," *mimeo, University of Tübingen*.

Englmaier, F., J. Galdon-Sanchez, R. Gil, M. Kaiser, and H. Strandt (2025): "Management practices and firm performance during the Great Recession: Evidence from Spanish survey data," *CEPR Discussion Paper No. 20749*.

EU Commission (2025): "Digital Decade 2025 Country Reports: Germany," *Staff Working Document 294 final*.

Federal Ministry of the Interior (2025): *Abschlussbericht zum Digitalisierungsprogramm OZG Föderal*, Bundesministerium des Innern, Berlin.

Fenizia, A. (2022): "Managers and productivity in the public sector," *Econometrica*, 90, 1063–1084.

——— (2024): "Regional Differences in Public-Sector Productivity and Managerial Talent," *AEA Papers and Proceedings*, 114, 592–596.

Hansen, S., M. McMahon, and A. Prat (2018): "Transparency and deliberation within the FOMC: A computational linguistics approach," *Quarterly Journal of Economics*, 133, 801–870.

Heldring, L. (2025): "The cost of state-building: Evidence from Germany," *NBER Working Paper No. 34586*.

Hjort, J., D. Moreira, G. Rao, and J. F. Santini (2021): "How research affects policy: Experimental evidence from 2,150 Brazilian municipalities," *American Economic Review*, 111, 1442–80.

Huettner, F. and M. Sunder (2012): "Axiomatic arguments for decomposing goodness of fit according to Shapley and Owen values," *Electronic Journal of Statistics*, 6, 1239–1250.

Ichniowski, C., K. Shaw, and G. Prenushi (1997): "The effects of human resource management practices on productivity," *American Economic Review*, 87, 291–313.

Israeli, O. (2007): "A Shapley-based decomposition of the R-square of a linear regression," *The Journal of Economic Inequality*, 5, 199–212.

Kunz, A. (2007): "Fusing Time and Space: The Historical Information System HGIS Germany," *International Journal of Humanities and Arts Computing*, 1, 111–122.

Leaver, C., O. Ozier, P. Serneels, and A. Zeitlin (2021): "Recruitment, effort, and retention effects of performance contracts for civil servants: Experimental evidence from Rwandan primary schools," *American Economic Review*, 111, 2213–46.

Lemos, R., K. Muralidharan, and D. Scur (2024): "Personnel management and school productivity: Evidence from india," *The Economic Journal*, 134, 2071–2100.

McCormack, J., C. Propper, and S. Smith (2014): "Herding cats? Management and university performance," *Economic Journal*, 124, F534–F564.

Normenkontrollrat (2024): "Gute Gesetze. Digitale Verwaltung. Weniger Bürokratie. Momentum nutzen, Wirkung steigern: NKR-Jahresbericht 2024," Nationaler Normenkontrollrat, Berlin.

Oster, E. (2019): "Unobservable selection and coefficient stability: Theory and evidence," *Journal of Business & Economic Statistics*, 37, 187–204.

Rasul, I. and D. Rogger (2018): "Management of bureaucrats and public service delivery: Evidence from the nigerian civil service," *Economic Journal*, 128, 413–446.

Rasul, I., D. Rogger, and M. J. Williams (2021): "Management, organizational performance, and task clarity: Evidence from ghana's civil service," *Journal of Public Administration Research and Theory*, 31, 259–277.

Ruge, K. and K. Ritgen (2021): "Local self-government and administration," in *Public Administration in Germany*, Palgrave Macmillan, Cham, 123–141.

Schrapper, L. (2021): "The Administration of the Länder," in *Public Administration in Germany*, Palgrave Macmillan, Cham, 105–121.

Scur, D., R. Sadun, J. Van Reenen, R. Lemos, and N. Bloom (2021): "The World Management Survey at 18: Lessons and the way forward," *Oxford Review of Economic Policy*, 37, 231–258.

Shapley, L. S. (1988): "A value for n-person games," in *The Shapley Value: Essays in Honor of Lloyd S. Shapley*, ed. by A. E. Roth, Cambridge: Cambridge University Press, 31–40.

Syverson, C. (2011): "What determines productivity?" *Journal of Economic Literature*, 49, 326–65.

Tirole, J. (2001): "Corporate Governance," *Econometrica*, 69, 1–35.

Weber, M. (1922): *Wirtschaft und Gesellschaft*, Mohr Siebeck, Tübingen.

Appendix

A Questionnaire sent to municipalities

In this Appendix we provide a translation of the survey questions (originally in German). In the list below, each question (Q 1 through Q 49) is accompanied by a comment field in square bracket, which is structured as follows:

1. "Type and result" identifies the question type by spelling out the possible answer options (separated by "|"), followed by the respective response frequencies (in percent) in the order of the answer options. Some questions are conditional in the sense that they are only reached if some prior question is answered with "yes", indicated in parentheses (see e.g., Q 2).

2. "Subscore" indicates whether the respective question is included in the management score, which is (i) not the case if the entry is "n.a.", and (ii) otherwise it is included in the management score (and in the stated subscore). As the score aims to measure the degree of structured management, "worst outcome" indicates the answer option receiving a score of zero. Each subscore is based on the following questions: goal setting (Q1-Q8), monitoring of service quality (Q9-Q16), organizational learning (Q17-Q22), structured processes (Q24-Q27), human resource management (Q23, Q30(i)-Q30(iii), Q31-Q41, Q44), and digitization (Q45(i)-Q45(v) and Q46(i)-Q46(v)).

3. The questionnaire contains various question types:

   (a) "yes | no", "Likert" (5-point scale), and "Interval in days" indicates that participants can select one answer option only (and if the percentage results do not add up to 100 this implies that some participants did not answer this question, see e.g., Q 3).

   (b) "multiple options" implies that participants are free to select any number of the available answer options (and as a consequence the percentage results generally do not add up to 100, see e.g., Q 4).

   (c) "number" allows participants to enter an integer.

   (d) "free text" solicits a free-form input from participants.

   (e) Questions Q 30, Q 45, Q 46 contain several subquestions, which each individually enter the management score. Therefore, the comment field provides the results separately for each subquestion.

List of questions

Q 1: Does your municipality have a mission statement that is directed at citizens? [Type and result: *yes | no = 26 | 74*. Subscore: *goal setting* (worst outcome = *no*)]

Q 2: Is the mission statement that is directed at citizens easily accessible for them (e.g., on the municipal webpage or newsletter)? [Type and result: *yes | no (conditional on Q1 = yes) = 84 | 16.* Subscore: *goal setting* (worst outcome = *no*)]

Q 3: Does the mission statement that is directed at citizens entail goals? [Type and result: *yes | no (conditional on Q1 = yes) = 86 | 12.* Subscore: *goal setting* (worst outcome = *no*)]

Q 4: What is the time horizon envisioned for these goals? [Type and result: *multiple options (conditional on Q1 = yes and Q3 = yes): one year | two to four years | five to ten years | more than 10 years = 7 | 21 | 62 | 31.* Subscore: *goal setting* (worst outcome = *no option chosen*)]

Q 5: Does your municipality have a mission statement that is specifically directed at the employees of the municipal administration? [Type and result: *yes | no = 21 | 78.* Subscore: *goal setting* (worst outcome = *no*)]

Q 6: Is the mission statement that is directed at employees easily accessible for them (e.g., on the municipal intranet)? [Type and result: *yes | no (conditional on Q5 = yes) = 88 | 11.* Subscore: *goal setting* (worst outcome = *no*)]

Q 7: Does the mission statement that is directed at employees entail goals? [Type and result: *yes | no (conditional on Q5 = yes) = 77 | 23.* Subscore: *goal setting* (worst outcome = *no*)]

Q 8: What is the time horizon envisioned for these goals? [Type and result: *multiple options (conditional on Q5 = yes and Q7 = yes): one year | two to four years | five to ten years | more than 10 years = 18 | 47 | 34 | 27.* Subscore: *goal setting* (worst outcome = *no option chosen*)]

Q 9: Does your municipality employ quantitative measures to evaluate administrative performance or service quality (e.g., with regard to processing times)? [Type and result: *yes | no = 16 | 83.* Subscore: *monitoring of service quality* (worst outcome = *no*)]

Q 10: Which of the following indicators are employed to measure administrative performance or service quality in your municipality? [Type and result: *multiple options (conditional on Q9 = yes): time to complete a process | waiting times of customers | frequency of complaints | other = 79 | 56 | 54 | 20.* Subscore: *monitoring of service quality* (worst outcome = *no option chosen*)]

Q 11: How are the indicators of administrative performance or service quality taken into account in municipal management? [Type and result: *multiple options (conditional on Q9 = yes): discussed at the employee level | at the management level | in the municipal council = 63 | 93 | 21.* Subscore: *monitoring of service quality* (worst outcome = *no option chosen*)]

Q 12: Overall, how strongly do such indicators affect the management of administrative performance or service quality in your municipality? [Type and result: *Likert (conditional on Q9 = yes): very large | large | medium | small | very small = 0 | 14 | 53 | 29 | 3.* Subscore: *monitoring of service quality* (worst outcome = *very small*)]

Q 13: Does your municipality conduct surveys among citizens and other customers to measure administrative performance or service quality? [Type and result: *yes | no = 16 | 84.* Subscore: *monitoring of service quality* (worst outcome = *no*)]

Q 14: How often does your municipality conduct surveys among citizens and other customers to measure administrative performance or service quality? [Type and result: *Likert (conditional on Q13 = yes): every year | every 2 years | every 5 years | every 10 years | less than every 10 years = 14 | 26 | 44 | 8 | 8.* Subscore: *monitoring of service quality* (worst outcome = *less than every 10 years*)]

Q 15: How are the results of surveys among citizens and other customers on performance or service quality taken into account in municipal management? [Type and result: *multiple options (conditional on Q13 = yes): discussed at the employee level | at the management level | in the municipal council = 62 | 87 | 25.* Subscore: *monitoring of service quality* (worst outcome = *no option chosen*)]

Q 16: Overall, how strongly do surveys among citizens and other customers affect the management of administrative performance or service quality in your municipality? [Type and result: *Likert (conditional on Q13 = yes): very large | large | medium | small | very small = 2 | 30 | 40 | 26 | 2.* Subscore: *monitoring of service quality* (worst outcome = *very small*)]

Q 17: In your municipality, is there a institutionalized way for employees to suggest improvements? [Type and result: *yes | no = 73 | 26.* Subscore: *organizational learning* (worst outcome = *no*)]

Q 18: Is there a reward system (financial or otherwise) for employees' suggestions for improvement in your municipal administration? [Type and result: *yes | no = 26 | 74.* Subscore: *organizational learning* (worst outcome = *no*)]

Q 19: How strongly do employees' suggestions for improvement affect municipal management? [Type and result: *Likert: very large | large | medium | small | very small = 3 | 26 | 34 | 25 | 12.* Subscore: *organizational learning* (worst outcome = *very small*)]

Q 20: Do employees who submit a suggestion for improvement usually receive feedback even if their proposal is ultimately not implemented? [Type and result: *yes | no = 91 | 8.* Subscore: *organizational learning* (worst outcome = *no*)]

Q 21: Is there a regular exchange between different departments of your municipal administration with the aim of improving processes or services? [Type and result: *yes | no = 72 | 27.* Subscore: *organizational learning* (worst outcome = *no*)]

Q 22: Does your municipality regularly exchange ideas with other municipalities on how to improve processes or services? [Type and result: *yes | no = 57 | 43.* Subscore: *organizational learning* (worst outcome = *no*)]

Q 23: What would happen in the case of staff shortages in key departments of your municipal administration? [Type and result: *multiple options: overtime | temporary relocation of idle capacity | hiring = 91 | 60 | 40.* Subscore: *human resource management* (worst outcome = *no option chosen*)]

Q 24: Does your municipal administration employ the instrument of "project management"? [Type and result: *yes | no = 49 | 51.* Subscore: *structured processes* (worst outcome = *no*)]

Q 25: How large is the value added of "project management" for your municipality? [Type and result: *Likert (conditional on Q24 = yes): very large | large | medium | small | very small = 7 | 51 | 34 | 7 | 0.* Subscore: *structured processes* (worst outcome = *very small*)]

Q 26: What role does the instrument of "process management" play in your municipal administration (e.g., the structured organization of processes in different departments)? [Type and result: *Likert: not used | few processes defined | core processes defined | processes defined in many areas | processes defined in all areas = 30 | 37 | 24 | 8 | 1.* Subscore: *structured processes* (worst outcome = *not used*)]

Q 27: Does your municipality use the instrument of a "product-oriented budgeting"? [Type and result: *yes | no = 79 | 20.* Subscore: *structured processes* (worst outcome = *no*)]

Q 28: To what extent do you agree with the following statement: The municipal administration and the municipal council work together in a constructive fashion? [Type and result: *Likert: very strongly | strongly | medium | weakly | very weakly = 19 | 39 | 30 | 9 | 3.* Subscore: *n.a.* (worst outcome = *n.a.*)]

Q 29: How many people are employed in your municipal administration (in full-time equivalents)? [Type and result: *number = 357 (mean), 80 (median).* Subscore: *n.a.* (worst outcome = *n.a.*)]

Q 30: How frequently does your municipal administration use the following instruments to remunerate its employees? [Type and result: *Likert: very often | often | sometimes | rarely | never. Three separate questions: (i) compensation above collective agreement, (ii) other upwards deviations from collective agreement, (iii) fast-track promotions) = (i) 1 | 5 | 19 | 32 | 39 , (ii) 0 | 2 | 12 | 28 | 51, (iii) 0 | 10 | 40 | 32 | 16.* Subscore: *human resource management* (worst outcome = *never*)]

Q 31: Does your municipal administration employ the instrument of "performance-based remuneration"? [Type and result: *yes | no = 78 | 22.* Subscore: *human resource management* (worst outcome = *no*)]

Q 32: To what extent do you agree with the following statement: In the municipal administration, the award of "performance-based remuneration" does indeed reflect actual performance differences? [Type and result: *Likert (conditional on Q31=yes): very strongly | strongly | medium | weakly | very weakly = 5 | 22 | 28 | 24 | 21.* Subscore: *human resource management* (worst outcome = *very weakly*)]

Q 33: Apart from "performance-based remuneration", what other instruments does your municipal administration use to reward high performance of employees? [Type and result: *multiple options: promotion | nonmonetary awards | rewards (e.g., vouchers) | work cell phone or tablet | training opportunities | other = 42 | 2 | 13 | 28 | 54 | 28.* Subscore: *human resource management* (worst outcome = *no option chosen*)]

Q 34: For promotion decisions in your municipal administration, how big a role does the seniority of eligible candidates play in practice? [Type and result: *Likert: very large | large | medium | small | very small = 0 | 8 | 31 | 36 | 25.* Subscore: *human resource management* (worst outcome = *very small*)]

Q 35: Before the onset of the COVID19 pandemic, how frequently did your municipal administration's employees "work from home"? [Type and result: *Likert: very often | often | sometimes | rarely | never = 2 | 6 | 20 | 54 | 18.* Subscore: *human resource management* (worst outcome = *never*)]

Q 36: Before the onset of the COVID19 pandemic, what challenges did your municipal administration encounter when aiming to implement "work from home"? [Type and result: *multiple options: insufficient technical equipment | lack of IT skills | data security concerns | hampered communication | lack of monitoring | unequal treatment of employees = 64 | 14 | 51 | 44 | 29 | 44.* Subscore: *human resource management* (worst outcome = *all options chosen*)]

Q 37: Before the onset of the COVID19 pandemic, which of the following events were usually organized in your municipal administration? [Type and result: *multiple options: christmas party | office party | honoring jubilees | other = 88 | 76 | 85 | 44.* Subscore: *human resource management* (worst outcome = *no option chosen*)]

Q 38: In your municipal administration, is there a collective agreement in place concerning individual "review meetings" with employees? [Type and result: *yes | no = 57 | 42.* Subscore: *human resource management* (worst outcome = *no*)]

Q 39: How frequently do individual "review meetings" with employees take place? [Type and result: *Likert: more than once per year | once per year | regularly but not every year | only when need arises | never = 11 | 58 | 10 | 18 | 3.* Subscore: *human resource management* (worst outcome = *never*)]

Q 40: In the individual "review meetings", do employees typically receive feedback on their performance? [Type and result: *yes | no (conditional on Q39 = yes) = 79 | 20.* Subscore: *human resource management* (worst outcome = *no*)]

Q 41: What would typically happen if employees in your municipal administration underperform over a prolonged period of time? [Type and result: *multiple options: individual review meeting | training | re-assignment | deadline for improvement | termination during probation period | other = 97 | 65 | 31 | 41 | 76 | 19.* Subscore: *human resource management* (worst outcome = *no option chosen*)]

Q 42: How many sick days do employees of the municipal administration approximately take (per employee and year, on average over the last few years)? [Type and result: *interval in days: 0 - 5 | 6 - 10 | 11 - 15 | 16 - 20 | 21 - 25 | 26 - 30 | 31 or more = 8 | 38 | 28 | 11 | 4 | 2 | 1.* Subscore: *n.a.* (worst outcome = *n.a.*)]

Q 43: When considering the annual average over the last few years: How many employees of the municipal administration have been on long-term sick leave such that a formal process of "re-integration" was triggered? [Type and result: *number = 17 (mean), 3 (median).* Subscore: *n.a.* (worst outcome = *n.a.*)]

Q 44: In times of scarce talent it is increasingly important to attract qualified personnel. What measures does your municipal administration take to achieve this? [Type and result: *multiple options: compensation above collective agreement | flexible work schedule | work cell phone or tablet | providing accommodation | support in finding accommodation | participation in job fairs | internship program | other = 15 | 80 | 41 | 2 | 29 | 29 | 31 | 39.* Subscore: *human resource management* (worst outcome = *no option chosen*)]

Q 45: In your municipal administration, approximately what percentage of relevant data is stored in digital form in the following areas? [Type and result: *Likert: 0-20% | 21-40% | 41-60% | 61-80% | 81-100%. Five separate questions: (i) personnel data, (ii) financial data, (iii) customer feedback, (iv) procurement processes, (v) processing times and status of processes = (i) 14 | 11 | 15 | 21 | 32, (ii) 2 | 4 | 8 | 21 | 57, (iii) 32 | 14 | 19 | 14 | 12, (iv) 9 | 16 | 23 | 22 | 23, (v) 43 | 17 | 14 | 10 | 8.* Subscore: *digitization* (worst outcome = *0-20%*)]

Q 46: How often does your municipal administration use data-based forecasting (e.g., statistical techniques that allow to make predictions) in the following areas? [Type and result: *Likert: very often | often | sometimes | rarely | never. Five separate questions: (i) customer services, (ii) financial planning, (iii) infrastructure planning, (iv) personnel planning, (v) childcare services = (i) 2 | 8 | 20 | 32 | 31, (ii) 18 | 40 | 19 | 8 | 7, (iii) 6 | 26 | 29 | 20 | 11, (iv) 5 | 26 | 26 | 21 | 15, (v) 12 | 27 | 21 | 17 | 15.* Subscore: *digitization* (worst outcome = *never*)]

Q 47: In your opinion, what are the biggest challenges for successful municipal management in the medium to long term? [Type and result: *free text*. Subscore: *n.a.* (worst outcome = *n.a.*)]

Q 48: What external conditions would have to change for municipal management to be better able to respond to pertinent challenges? [Type and result: *free text*. Subscore: *n.a.* (worst outcome = *n.a.*)]

Q 49: In your opinion, what hampers an effective and timely expansion of e-government services? [Type and result: *free text*. Subscore: *n.a.* (worst outcome = *n.a.*)]

B Additional tables and figures

B.1 Regional types and population size brackets: Distributions

Table 8: Regional and population types: Descriptive statistics for participating and non-participating municipalities

	Non-Participants	Participants	Total
Regional type			
Metropolitan	624 (27.05%)	162 (27.00%)	786 (27.04%)
Urban	465 (20.16%)	124 (20.67%)	589 (20.26%)
Rural close to urban	619 (26.83%)	171 (28.50%)	790 (27.18%)
Rural	599 (25.96%)	143 (32%)	742 (25.52%)
Total	2,307 (100.00%)	600 (100.00%)	2,907 (100.00%)
Population size bracket			
2,001 to 5,000	40 (1.62%)	17 (2.83%)	57 (1.85%)
5,001 to 20,000	1844 (74.47%)	426 (71.00%)	2,270 (73.80%)
20,001 to 50,000	447 (18.05%)	111 (18.50%)	558 (18.14%)
50.001 to 100,000	85 (3.43%)	25 (4.17%)	110 (3.58%)
100,001 to 500,000	49 (1.98%)	18 (3.00%)	67 (2.18%)
more than 500,000	11 (.44%)	3 (.50%)	14 (.46%)
Total	2,476 (100.00%)	600 (100.00%)	3,076 (100.00%)

Notes: The table displays the distributions of participating and non-participating municipalities across regional types and population size brackets. The population size bracket to which a given municipality is assigned is time-invariant across our sample period 2015-2019 for all but seven of the 600 participating municipalities. To these seven municipalities, we assign the 2015 population size bracket.

B.2 Management scores of private firms in Germany

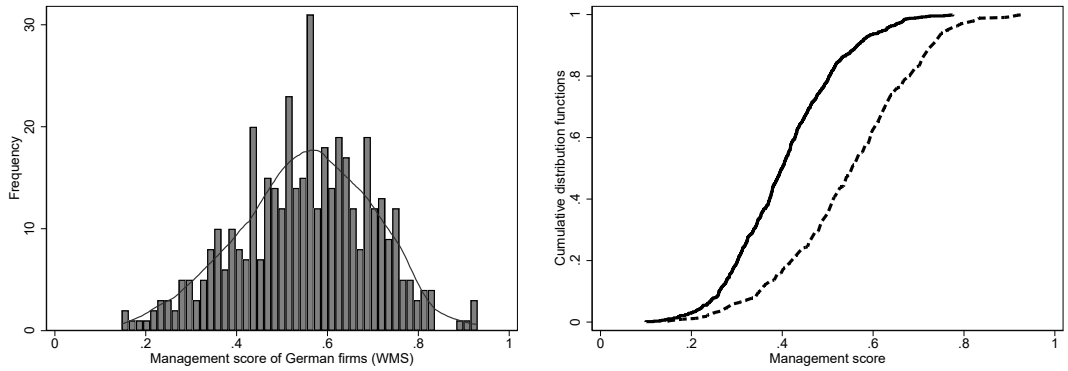
In this appendix, we construct a management score distribution for a sample of private-sector firms in Germany. As explained in the main text, this serves as a tentative benchmark to compare our results on municipal administrations to private-sector firms. On the website accompanying their textbook, Békés and Kézdi (2021a) provide World Management Survey data, which also includes manufacturing firms that are based in Germany (in particular, see Békés and Kézdi, 2021b). We retrieved these data from https://osf.io/t6zdp/overview on October 11, 2026.

The data is based on the survey used in Bloom and Van Reenen (2007) containing 18 questions covering performance measurement, goal setting, operations, and human resource management policies. The answers to each of the 18 questions are coded on a scale from 1 to 5 points, and each firm's management score is calculated as the average number of points obtained (resulting in scores between 1 and 5). In contrast, for the municipal scores, we follow Bloom et al. (2019) by normalizing the responses to each question to values between 0 and 1 (see Section 4.2). Consequently, municipal management scores take values between 0 and 1. Hence, for the sake of comparability, we transform the management scores of Békés and Kézdi (2021b) using $s_T = (s-1)/4$, where s is the original score, and s_T the transformed value. This transformation yields management scores for firms that also lie between 0 and 1.

The data contain 474 observations of firms in Germany, where some firms were surveyed more than once. In such cases, we take the most recent management score observation, yielding a total of 424 firm observations from 2004 to 2014.

The left panel of Figure 7 shows the frequency distribution of the (transformed) private-sector management scores. The right panel shows the cumulative distribution functions of the management scores of municipalities and of private-sector firms, revealing that the distribution of private-sector scores first-order stochastically dominates that of municipalities.

Figure 7: Frequency distribution of the management score of private-sector firms in Germany (left) and cumulative distribution functions of the management scores of municipal administrations (bold line) and private-sector firms (dotted line) in Germany (right)



Notes: In the left panel, $N = 424$, mean score $= .55$, standard deviation $= .15$, 25%-percentile $= .46$, 50%-percentile $= .56$, 75%-percentile $= .65$. The black line indicates the corresponding (Epanechnikov) kernel density.

B.3 Robustness: Management and performance

Table 9: Robustness: Management score and municipal performance (3-year- averages: 2017-2019)

	(1)	(2)	(3)	(4)
Net migration	5.37**	5.58***	5.12***	4.56**
	(0.02)	(0.00)	(0.00)	(0.02)
Academic workforce	12.10***	12.00***	10.42***	4.61
	(0.00)	(0.00)	(0.00)	(0.18)
Start-ups	1.61***	1.22***	0.94**	0.49
	(0.00)	(0.00)	(0.01)	(0.26)
Broadband coverage	27.41***	28.27***	25.31***	11.34
	(0.00)	(0.00)	(0.00)	(0.16)
Business tax base	128.19***	150.05**	146.59**	125.88
	(0.00)	(0.01)	(0.01)	(0.13)
Income tax revenue	70.36	80.11**	36.19	57.25***
	(0.17)	(0.01)	(0.19)	(0.01)
Primary balance	32.98	97.44	98.77	87.42
	(0.54)	(0.34)	(0.32)	(0.44)
Financial balance	30.53***	28.30***	28.65***	12.78
	(0.00)	(0.00)	(0.00)	(0.25)
Municipal performance index (MPI)	1.90***	1.97***	1.75***	1.10**
	(0.00)	(0.00)	(0.00)	(0.04)
State dummies	No	Yes	Yes	Yes
Regional type dummies	No	No	Yes	Yes
Population bracket dummies	No	No	No	Yes

Notes: The note below Table 3 applies with the only difference that three-year averages (2017-2019) of the indicators of municipal performance are considered.

Table 10: Robustness: Management score and municipal performance (2019 only)

	(1)	(2)	(3)	(4)
Net migration	3.49	3.78	3.21	2.68
	(0.12)	(0.11)	(0.15)	(0.28)
Academic workforce	12.75***	12.65***	11.07***	4.87
	(0.00)	(0.00)	(0.00)	(0.23)
Start-ups	1.85***	1.05	0.81	0.41
	(0.00)	(0.12)	(0.24)	(0.70)
Broadband coverage	22.44***	23.18***	20.60***	10.07
	(0.00)	(0.00)	(0.00)	(0.13)
Business tax base	122.65***	155.93**	151.11**	139.09*
	(0.00)	(0.03)	(0.04)	(0.10)
Income tax revenue	70.09	79.30**	35.42	61.67**
	(0.18)	(0.02)	(0.20)	(0.02)
Primary balance	85.52	149.04	136.91	130.96
	(0.48)	(0.32)	(0.34)	(0.46)
Financial balance	28.98***	25.66**	25.87***	10.07
	(0.00)	(0.02)	(0.01)	(0.46)
Municipal performance index (MPI)	1.78***	1.83***	1.59***	1.01*
	(0.00)	(0.00)	(0.00)	(0.08)
State dummies	No	Yes	Yes	Yes
Regional type dummies	No	No	Yes	Yes
Population bracket dummies	No	No	No	Yes

Notes: The note below Table 3 applies with the only difference that the most recent (i.e. 2019) values of the indicators of municipal performance are considered.

Table 11: Robustness: z-score and municipal performance

	(1)	(2)	(3)	(4)
Net migration	0.87***	0.84***	0.78***	0.69***
	(0.00)	(0.00)	(0.00)	(0.00)
Academic workforce	1.35***	1.32***	1.15***	0.48
	(0.00)	(0.00)	(0.00)	(0.20)
Start-ups	0.21***	0.16***	0.12***	0.07
	(0.00)	(0.01)	(0.01)	(0.17)
Broadband coverage	3.08***	3.06***	2.72***	1.21
	(0.00)	(0.00)	(0.00)	(0.13)
Business tax base	15.53***	17.34***	16.83**	14.29
	(0.00)	(0.00)	(0.01)	(0.11)
Income tax revenue	10.61**	10.12***	5.06	6.83**
	(0.05)	(0.00)	(0.10)	(0.01)
Primary balance	11.46	16.19*	16.03*	16.47
	(0.14)	(0.09)	(0.09)	(0.15)
Financial balance	4.65**	4.54**	4.52***	2.79
	(0.03)	(0.04)	(0.00)	(0.12)
Municipal performance index (MPI)	0.25***	0.25***	0.22***	0.15**
	(0.00)	(0.00)	(0.00)	(0.04)
State dummies	No	Yes	Yes	Yes
Regional type dummies	No	No	Yes	Yes
Population bracket dummies	No	No	No	Yes

Notes: The note below Table 3 applies with the only difference that the z-score is used as the measure of structured management (instead of the management score). The construction of the z-score follows Bloom et al. (2019): The answers to each of the 52 questions entering the score are normalized to have a mean of 0 and a standard deviation of 1, and then averages are taken over all questions.

Table 12: Robustness: Management score and municipal performance (bounding exercise of Oster (2019) to assess potential omitted variable bias)

Performance Indicator	$R^2_{(1)}$	$R^2_{(4)}$	δ	β
Net migration	0.01	0.17	5.67	5.32
Academic workforce	0.07	0.37	1.31	1.07
Start-ups	0.01	0.34	1.10	0.05
Broadband coverage	0.03	0.29	1.71	5.11
Business tax base	0.02	0.09	6.52	113.68
Income tax revenue	0.00	0.75	5.24	44.98
Primary balance	0.00	0.32	-9.54	149.10
Financial balance	0.02	0.22	2.58	13.91
Municipal performance index (MPI)	0.06	0.44	2.88	0.88

Notes: This table shows the results of the Oster (2019) bounding exercise for model specifications (1) and (4) for each of the eight performance indicators and the municipal performance index (MPI) as reported in Table 3 (Stata package *psacalc*). In column 2, $R^2_{(1)}$ is the R^2 from (the restricted) model (1) without controls. In column 3, $R^2_{(4)}$ is the R^2 from (the full) model (4) which controls for state, regional type and population size. In column 4, δ is the degree of proportional selection on unobservables relative to observables required to reduce the treatment effect to zero. In column 5, β is the bias-adjusted treatment effect assuming $\delta = 1$. As suggested in Oster (2019), in all specifications the assumed maximal R^2 is 1.3 times the R^2 of the respective full model (4).

Table 13: Robustness: Management style and municipal performance

	(1)	(2)	(3)	(4)
Net migration	1.83**	1.82***	1.58***	1.19*
	(0.03)	(0.00)	(0.01)	(0.06)
Academic workforce	4.31***	4.42***	3.73***	1.21
	(0.00)	(0.00)	(0.00)	(0.15)
Start-ups	0.56***	0.49***	0.37*	0.19
	(0.00)	(0.00)	(0.05)	(0.31)
Broadband coverage	9.35***	9.41***	8.27***	2.51
	(0.00)	(0.00)	(0.00)	(0.20)
Business tax base	45.83**	51.95**	50.02**	41.15
	(0.02)	(0.01)	(0.02)	(0.19)
Income tax revenue	13.94	24.66**	4.90	10.18
	(0.25)	(0.02)	(0.50)	(0.19)
Primary balance	33.53	55.30	55.25	59.31
	(0.24)	(0.11)	(0.12)	(0.19)
Financial balance	14.76***	13.76***	14.06***	7.63
	(0.00)	(0.00)	(0.00)	(0.26)
Municipal performance index (MPI)	0.71***	0.75***	0.65***	0.38*
	(0.00)	(0.00)	(0.00)	(0.08)
State dummies	No	Yes	Yes	Yes
Regional type dummies	No	No	Yes	Yes
Population bracket dummies	No	No	No	Yes

Notes: The note below Table 3 applies with the only difference that the probability for the Type 1 management style is used as the measure of structured management instead of the management score (see Section 6.3).

B.4 Explaining the management score

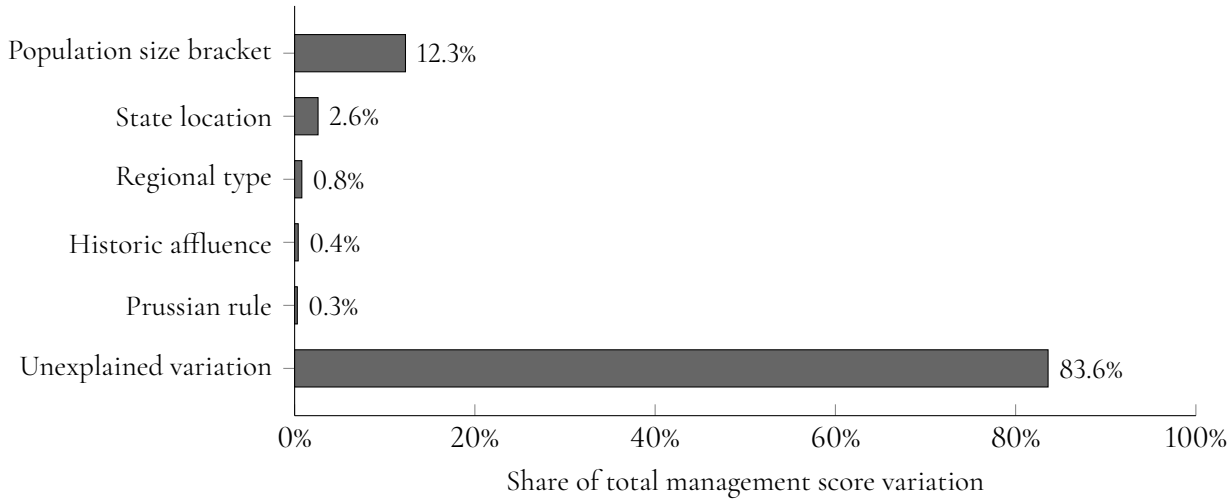
This appendix investigates potential sources of the variation in management scores: municipalities' state location, regional type, and population size bracket (as introduced in Section 4), historic affluence, and having been subject to Prussian rule in the past.

The rationale for considering historic affluence is that municipalities that were relatively affluent in the past might exhibit a higher management score today (e.g. because they had sufficient financial resources to systematically evaluate, refine, or change their management policies). We approximate a municipality's historic affluence by the variable *Historic primary balance*, defined as the average value of the variable *Primary balance* (a key measure of municipalities' financial health as defined in Section 5.2 above) in the years 2006 and 2007, the earliest dates available in our data set.

We also consider the more distant past, specifically whether a municipality has historically been under Prussian rule. Prussia is widely noted for its early establishment of formal bureaucratic structures and local state capacity in Germany (see e.g., Heldring, 2025). Testing for historical Prussian influence allows us to assess whether modern municipal management reflects long-standing institutional path dependency. We follow Heldring (2025) in classifying Prussian affiliation based on municipal boundaries from the HGIS Germany spatial database (Kunz, 2007), and in adopting 1890 as the reference cutoff year.

In a first step, we regress the management score on the variables discussed above. This results in $R^2 = 0.151$. In a second step, we perform a Shapley value decomposition (as explained in Section 6.2) to determine these variables' shares in the explained variation. As shown in Figure 8, the population size has the largest share, but still accounts for only slightly above 12 percent of the management score variation. The impact of all other variables is even substantially lower.

Figure 8: Explaining the management score variation: Shapley value decomposition



Note: Based on the Shapley value decomposition, the figure shows the different variables' share in explaining the overall management score variation. For example, based on $R^2 = 0.151$, the decomposition yields 0.73 for the population size bracket, and hence a Shapley value of $0.73 \times 0.151 = 0.11$. That is, population size accounts for 11 percent of the overall management score variation.